

# Analysis of the performance of kinetic reaction mechanisms in estimating N<sub>2</sub>O mole fractions in 70/30 vol% NH<sub>3</sub>/H<sub>2</sub> premixed flames

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## ARTICLE INFO

### Keywords:

Kinetic reaction mechanism  
Ammonia  
Burner stabilised-stagnation flow  
Kinetic modelling  
N<sub>2</sub>O mole fraction

## ABSTRACT

To address the environmental concerns associated with fossil fuels, this study explores ammonia (NH<sub>3</sub>) blended with hydrogen (H<sub>2</sub>) as an alternative fuel. While offering reduced CO<sub>2</sub> emissions and leveraging existing infrastructure, NH<sub>3</sub>-H<sub>2</sub> combustion notably leads to the production of nitrogen oxides (NO<sub>x</sub>), including the greenhouse gas nitrous oxide (N<sub>2</sub>O). Understanding the flame structure and chemistry responsible for N<sub>2</sub>O formation and consumption is crucial. This study comprehensively investigates various kinetic reaction mechanisms, focusing on accurately estimating N<sub>2</sub>O mole fractions and identifying mechanisms that align closely with experimental data. Sixty-seven chemical kinetic mechanisms have been numerically analysed across various equivalence ratios ( $\phi$ ) ranging from 0.57 to 1.4, utilizing the Premixed Stagnation Flame Model via Chemkin-Pro software. Simulations in a Perfectly Stirred Reactor were also conducted for the kinetic models that demonstrated high accuracy within the burner-stabilized stagnation flame model and closely matched experimental measurements with minimal discrepancies. This was done to determine whether the tested models, which perform accurately, maintain their performance across different combustion configurations. A preliminary assessment was carried out using the Normalized Error Approach, taking into account the uncertainty of experimental measurements, to compare the numerical results with experimental data. This method is significant in determining whether the discrepancies between the model's calculations and the experimental results, considering the experimental uncertainties, are within an acceptable range of error. The sensitivity analysis along with rate of production/consumption of N<sub>2</sub>O investigation at several conditions of equivalence ratio (0.6, 1, 1.4) has been conducted to check the discrepancies among the mechanisms and shed light on the reactions that dominate the formation/consumption of N<sub>2</sub>O at different conditions. The study revealed that the kinetic model developed by Klippenstein et al. (2018) demonstrates remarkable accuracy in predicting N<sub>2</sub>O mole fractions across a range of conditions, specifically within the equivalence ratio range of 0.6 to 1.4. In this range, the normalised error values were observed to be less than 1, signifying that the experimental values align closely with the numerical expectations, considering the uncertainty. However, it is noteworthy that the model's accuracy appears to decrease in lean flame scenarios, particularly when the equivalence ratio falls between 0.57 and 0.585. In these conditions, higher normalised error values exceeding 1 were recorded, suggesting a possible deviation between numerical predictions and experimental observations. Along with that the rate of production/consumption analysis revealed the NH + NO  $\rightleftharpoons$  N<sub>2</sub>O + H reaction has a dominant role in the formation of N<sub>2</sub>O for all studied conditions, while the consumption of N<sub>2</sub>O is dominated by reactions N<sub>2</sub>O + H  $\rightleftharpoons$  N<sub>2</sub> + OH and N<sub>2</sub>O (+M)  $\rightleftharpoons$  N<sub>2</sub> + O (+M) at all investigated conditions.

## 1. Introduction

The vast spreading population and rapid economic development in

recent decades have affected dramatically the global energy consumption. Fossil fuel sources such as coal, petroleum and natural gas are kept at the top of the major energy sources across the world. The emissions

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<https://doi.org/10.1016/j.fuel.2024.131897>

Received 18 September 2023; Received in revised form 16 April 2024; Accepted 12 May 2024

Available online 24 May 2024

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**Table 1**  
Boundary conditions used in the experiments [19].

| #  | Equivalence ratio $\varphi$ | $V_{in}$ (cm/s) | Plate temperature $T_W$ (K) |
|----|-----------------------------|-----------------|-----------------------------|
| 1  | 0.57                        | 23.4            | 489.7                       |
| 2  | 0.585                       | 24.26           | 491.3                       |
| 3  | 0.6                         | 25.53           | 493.5                       |
| 4  | 0.65                        | 27.09           | 496.6                       |
| 5  | 0.7                         | 27.93           | 499.2                       |
| 6  | 0.75                        | 29.18           | 503.8                       |
| 7  | 0.8                         | 31.26           | 511.5                       |
| 8  | 1.0                         | 42.57           | 563.6                       |
| 9  | 1.2                         | 40.96           | 574.7                       |
| 10 | 1.4                         | 30.86           | 504.0                       |

and the pollutants that can be released by the combustion of these fossil fuels include CO<sub>2</sub>, CO, NO<sub>x</sub>, SO<sub>2</sub>, volatile compounds, particulate matter, etc. which have extremely negative impacts on the ecosystem [1]. Due to the negative effects of CO<sub>2</sub>, it has been necessary to develop new technologies that aim to reduce problems that correlate with energy consumption. These factors combined with more strict regulations lead to investigate carbon-free fuel sources linked to renewable energy resources [2]. Ammonia (NH<sub>3</sub>) is a promising alternative carbonless fuel due its high hydrogen density content, hence making it an attractive hydrogen carrier fuel. The fuel enables 1) CO<sub>2</sub>, SO<sub>x</sub> and soot emission free flue gases; 2) producibility from different sources such as renewable sources and biomass; 3) transportation and storage can be done using working existing infrastructure. All these factors make NH<sub>3</sub> a favourably

clean fuel candidate for the energy sector. However, high NO<sub>x</sub> emission and narrow flammability limits are the main problem that restricts the use of NH<sub>3</sub> in large-scale thermal devices [3,4]. The combustion properties of NH<sub>3</sub>, in terms of laminar burning velocity, have been studied and improved by blending NH<sub>3</sub> with other doping agents as a fuel. However, this can increase NO<sub>x</sub> emissions [3–6].

The term NO<sub>x</sub> stands for all nitrogen oxide forms generated by combustion, which are mainly nitric oxide (NO), nitrogen dioxide (NO<sub>2</sub>) and nitrous oxide (N<sub>2</sub>O). The greenhouse effect of N<sub>2</sub>O is also very important as it has 300 times larger Global Warming Potential than CO<sub>2</sub> [7,8]. Several experimental and numerical studies have been carried out on NH<sub>3</sub>-H<sub>2</sub> blends in terms of N<sub>2</sub>O emission [9–11]. A study on turbulent swirl-stabilized ammonia/hydrogen flames using a tangential swirl burner by Alnasif et al. [9] identified that the concentration of N<sub>2</sub>O peaks at an 85/15 vol% NH<sub>3</sub>/H<sub>2</sub> blend with a thermal power of 20 kW and a Reynolds number of 40,000. Another study conducted by Okafor et al. [12] on turbulent, swirl-stabilized ammonia/air flames indicated that heat loss promotes N<sub>2</sub>O emissions. However, due to the complex flow structure within a swirl combustor, it is challenging to analyse the production and reduction mechanisms of N<sub>2</sub>O, making this study unavailable for chemistry validation. Numerically, many studies have been carried out on NH<sub>3</sub> in terms of understanding the chemical kinetics of NH<sub>3</sub> combustion to improve the reaction mechanism of NO<sub>x</sub>. The performance of most advanced kinetic models was improved on the basis of experiments conducted on many combustion configurations, an exercise that resulted in adding new reactions or updating rate parameters that govern the rate of reactions [13–16].

**Table 2**  
Kinetic reaction mechanisms adopted in the present study.

| #  | Kinetic mechanism            | No. of reactions | No. of species | Ref. | #  | Kinetic mechanism                              | No. of reactions | No. of species | Ref. |
|----|------------------------------|------------------|----------------|------|----|------------------------------------------------|------------------|----------------|------|
| 1  | Bertolino et al., 2021       | 264              | 38             | [23] | 35 | Dagaut et al., 2008                            | 250              | 41             | [24] |
| 2  | Mei et al., 2021             | 264              | 38             | [25] | 36 | Gregory P. Smith et al., 2000                  | 325              | 53             | [26] |
| 3  | Han et al., 2021             | 298              | 36             | [27] | 37 | Coda Zabetta and Hupa, 2008                    | 371              | 60             | [28] |
| 4  | Tang, et al., 2022           | 211              | 35             | [29] | 38 | Alzueta MU, 2016                               | 654              | 131            | [30] |
| 5  | Gotama et al., 2022          | 119              | 26             | [13] | 39 | Shmakov et al., 2010                           | 1207             | 127            | [31] |
| 6  | Shrestha et al., 2021        | 1099             | 125            | [32] | 40 | Esarte et al., 2011                            | 536              | 79             | [33] |
| 7  | Z. Wang et al., 2021         | 444              | 91             | [34] | 41 | Abian et al., 2015                             | 201              | 31             | [35] |
| 8  | X. Zhang et al., 2021        | 263              | 38             | [18] | 42 | T. Wang et al., 2018                           | 925              | 81             | [36] |
| 9  | Arunthanayothin et al., 2021 | 2444             | 157            | [37] | 43 | T. Faravelli, 2017                             | 158              | 29             | [38] |
| 10 | Stagni et al., 2020          | 203              | 31             | [16] | 44 | POLIMI, 2014                                   | 155              | 29             | [39] |
| 11 | Han et al., 2019             | 177              | 35             | [40] | 45 | Marques et al., 2009                           | 318              | 61             | [41] |
| 12 | De Persis et al., 2020       | 647              | 103            | [42] | 46 | Aranda et al., 2013                            | 566              | 95             | [43] |
| 13 | Mei et al., 2019             | 265              | 38             | [44] | 47 | Jiang et al., 2020                             | 60               | 19             | [45] |
| 14 | Li et al., 2019              | 957              | 128            | [46] | 48 | Sun et al., 2022                               | 486              | 66             | [47] |
| 15 | Okafor et al., 2019          | 356              | 59             | [48] | 49 | Song et al., 2019                              | 158              | 29             | [49] |
| 16 | Glarborg et al., 2018        | 231              | 39             | [50] | 50 | Mével et al., 2009                             | 203              | 32             | [51] |
| 17 | Shrestha et al., 2018        | 1081             | 124            | [52] | 51 | Da Rocha et al., 2019 (Improved Mathieu mech.) | 66               | 22             | [53] |
| 18 | Otomo et al., 2018           | 213              | 32             | [54] | 52 | Da Rocha et al., 2019 (Improved Otomo mech.)   | 51               | 21             | [53] |
| 19 | U. Mechanism, 2018           | 41               | 20             | [55] | 53 | Da Rocha et al., 2019 (Improved Okafor mech.)  | 70               | 24             | [53] |
| 20 | Klippenstein et al., 2018    | 211              | 33             | [56] | 54 | Kovaleva et al., 2022                          | 354              | 59             | [57] |
| 21 | Nakamura et al., 2017        | 232              | 33             | [14] | 55 | Houshfar et al., 2012 (Midd temp.)             | 91               | 26             | [58] |
| 22 | Y. Zhang et al., 2017        | 251              | 44             | [59] | 56 | Houshfar et al., 2012 (High temp.)             | 430              | 52             | [58] |
| 23 | Lamoureux et al., 2016       | 934              | 123            | [60] | 57 | Houshfar et al., 2012 (Low temp.)              | 198              | 35             | [58] |
| 24 | Xiao et al., 2017            | 276              | 55             | [61] | 58 | Capriolo et al., 2021                          | 2300             | 201            | [62] |
| 25 | Song et al., 2016            | 204              | 32             | [63] | 59 | Xu et al., 2023                                | 389              | 69             | [64] |
| 26 | Nozari and Karabeyoglu, 2015 | 91               | 21             | [65] | 60 | Thomas et al., 2022                            | 1099             | 125            | [66] |
| 27 | Mathieu and Petersen, 2015   | 278              | 54             | [67] | 61 | Kovács et al., 2020                            | 214              | 34             | [68] |
| 28 | Duynslaegher et al., 2012    | 80               | 19             | [69] | 62 | Kovács et al., 2021                            | 537              | 70             | [70] |
| 29 | Klippenstein et al., 2011    | 202              | 31             | [71] | 63 | Saxena and Williams, 2007                      | 288              | 59             | [72] |
| 30 | K. Zhang et al., 2011        | 701              | 88             | [73] | 64 | Valkó et al., 2022                             | 537              | 70             | [74] |
| 31 | Lamoureux et al., 2010       | 883              | 119            | [75] | 65 | Alzueta et al., 2001                           | 464              | 65             | [76] |
| 32 | Konnov, 2009                 | 1207             | 127            | [77] | 66 | Nakamura and Shindo, 2019                      | 485              | 66             | [78] |
| 33 | Mendiara and Glarborg, 2009  | 779              | 79             | [79] | 67 | Glarborg, 2022                                 | 270              | 41             | [15] |
| 34 | Tian et al., 2009            | 703              | 84             | [80] |    |                                                |                  |                |      |

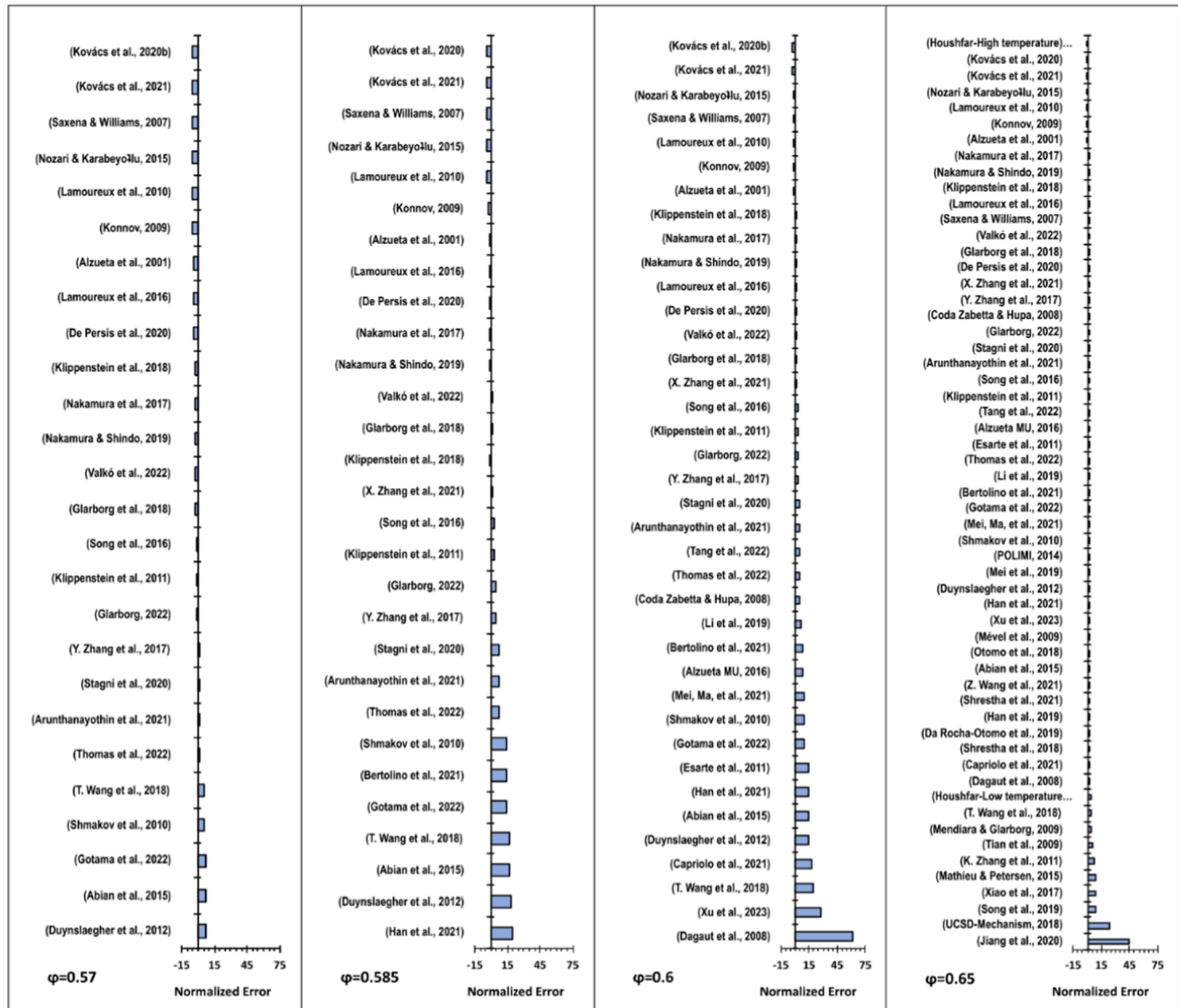


Fig. 1. Normalized error of  $N_2O$  mole fractions for 67 reaction mechanisms in lean  $NH_3/H_2$  flames, analysed using BSSF numerical data across equivalence ratios of 0.57 to 0.65. Notably, at conditions of 0.57, 0.585, and 0.6, numerous tested kinetic models failed to predict the  $N_2O$  mole fractions.

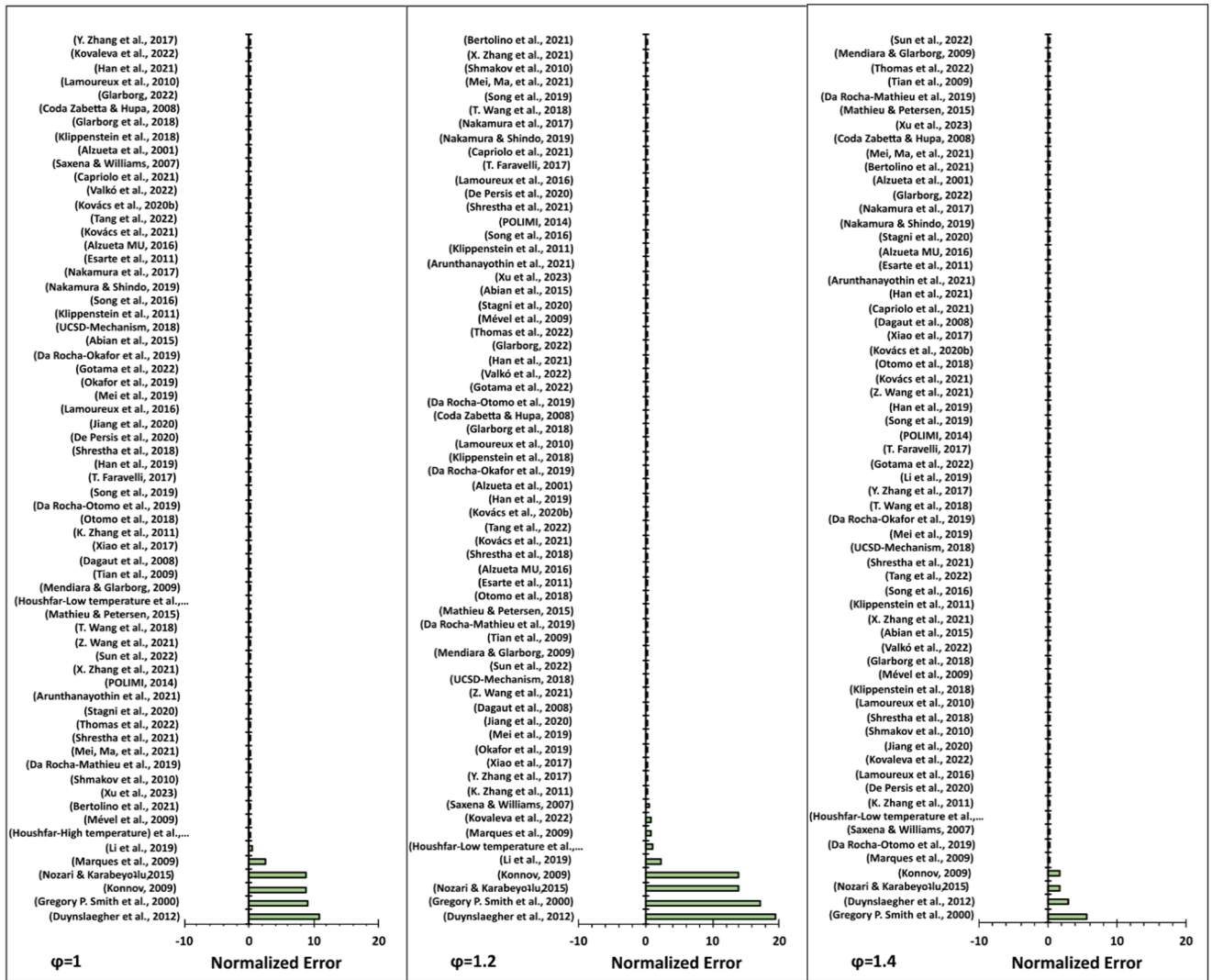


Fig. 2. Normalized Error of  $N_2O$  Mole Fractions for 67 Reaction Mechanisms in stoichiometry and rich  $NH_3/H_2$  Flames, Analysed Using BSSF Numerical Data Across Equivalence Ratios of 1 to 1.4.

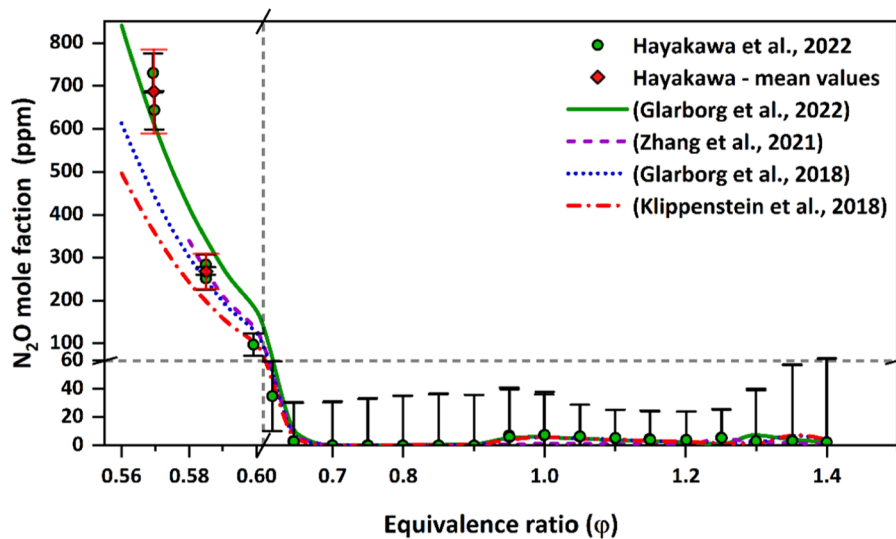


Fig. 3. Variation of  $N_2O$  mole fraction as a function of equivalence ratio for 70/30 vol%  $NH_3/H_2$  premixed flames using the Burner Stabilized Stagnation Flame Model (BSSF). The symbols denote experimental data, while the dotted lines represent simulation results. Scale breaks, shown as dashed grey lines, were introduced to show data variations across all scales clearly.

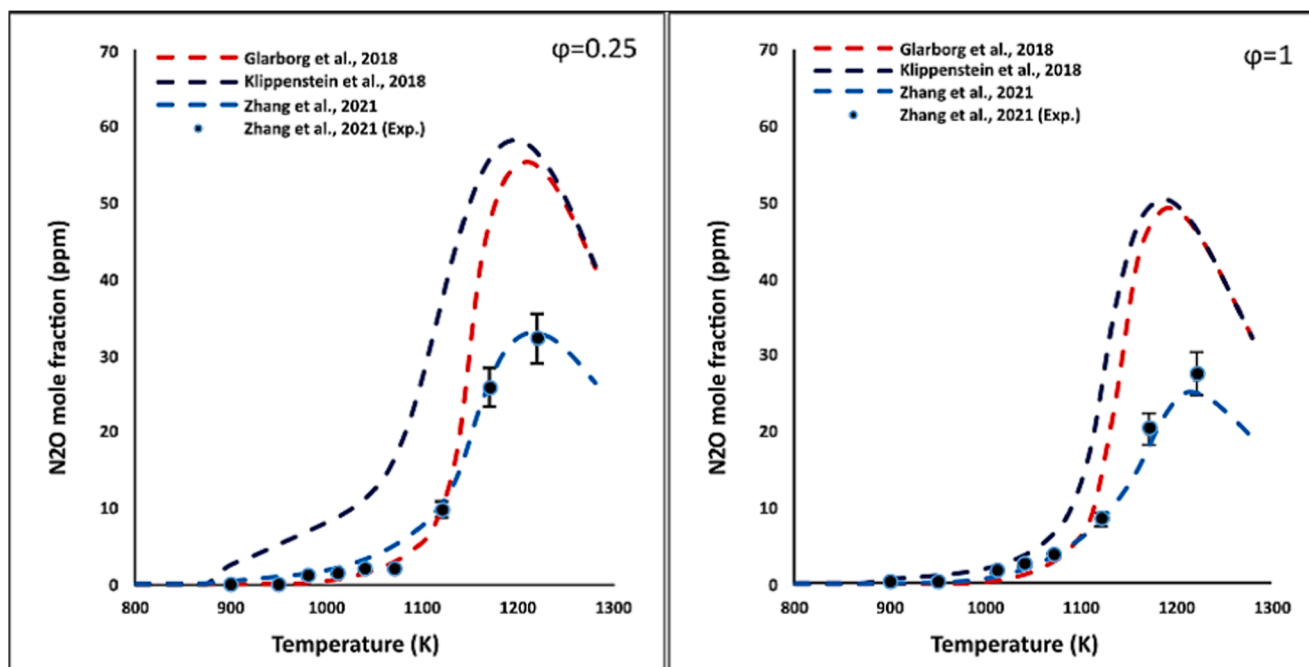


Fig. 4. Variation of  $\text{N}_2\text{O}$  mole fractions with temperature at equivalence ratios of 0.25 and 1. Experimental results are depicted by symbols, while simulations are illustrated with dotted lines. The model from Zhang et al. [18] has been fitted based on data reported in their study, explaining the close match observed above.

According to studies by Valera Medina et al. [3] and Mørch et al. [17], a 70/30 vol%  $\text{NH}_3/\text{H}_2$  blended fuel provides stable performance in fuelling gas turbine combustors. This blend has been selected for consideration in the present study. The literature on the experimental investigation of ammonia combustion has been extensively reviewed. To the best of our knowledge, the data chosen for validation, published by Zhang et al. [18] and Hayakawa et al. [19], is the only dataset found in the literature for  $\text{N}_2\text{O}$  speciation of a 70/30 vol%  $\text{NH}_3/\text{H}_2$  blend at atmospheric conditions and various equivalence ratios. In the study by Zhang et al. [18],  $\text{NH}_3/\text{H}_2$  mixtures were investigated using jet-stirred reactor (JSR) oxidation experiments at atmospheric pressure, covering a temperature range of 800–1280 K and two equivalence ratios (0.25 and 1). The data collected from these experiments helped refine their kinetic model, improving its ability to accurately predict the mole fractions of NO and  $\text{N}_2\text{O}$  under the conditions tested. Hayakawa and coworkers used the same system configuration (stabilized-stagnation flame configuration) as Brackmann et al. [20]. This configuration allows the flame to be stabilized without a pilot flame or the addition of chemical species to enhance flame intensity, which is a beneficial strategy for ammonia flames in terms of: 1) Temperature Distribution: The stagnation point heats up due to gas compression, aiding complete ammonia combustion. 2) Uniformity: Symmetrical gas flow creates even temperature and gas concentration, ensuring consistent product gases. 3) Stability: The steady gas flow at the stagnation point maintains a stable flame, which is less vulnerable to disturbances, ensuring consistent production of combustion products. These factors can influence combustion efficiency and NO<sub>x</sub> formation accuracy, reducing the amount of unburned fuel in the product gases. Furthermore, a stagnation-stabilized flame configuration can be easily simulated using CHEMKIN software [21] by applying the appropriate boundary conditions.

Given the aforementioned, selecting the dataset provided by Hayakawa et al. [19] for validation is advantageous. The only issue is the large uncertainty bars, which can be attributed to the FTIR gas analyser used for collecting gas products. This analyser is designed to measure up to 1000 ppmv, and when measuring  $\text{N}_2\text{O}$  at this upper limit, the results are less precise, with a significant margin of error or uncertainty, as mentioned by the group in another published work on ammonia flames

[22]. The current study aims to analyse the performance of kinetic reaction mechanisms for estimating  $\text{N}_2\text{O}$  in a 70/30 vol%  $\text{NH}_3/\text{H}_2$  blended fuel at the full range of equivalence ratios (0.57–1.4). Additionally, the study aims to identify the reaction steps responsible for  $\text{N}_2\text{O}$  formation/consumption and reveal the reasons for discrepancies in the estimation of experimental measurements of  $\text{N}_2\text{O}$  in ammonia combustion systems.

## 2. Numerical setup and kinetic modeling

Chemkin-Pro package of ANSYS software was used to study the performance of 67 chemical kinetic mechanisms in stabilised-stagnation flame simulations (BSSF). The numerical simulations applied the same boundary conditions as those used in the experiments in terms of atmospheric conditions, plate temperature and inlet velocity of the blend, Table 1. Also, the length of the computational domain was set to 2 cm in accordance with the distance of the top plate from the nozzle burner used in the experiments. In addition, the Soret effect (Thermal diffusion effect) was considered in this study due to the exothermic nature of combustion reactions. Further, the maximum number of grid points allowed with adaptive grid control, gradient and curvature thresholds were set to 5000, 0.01, 0.01, respectively.

Table 2 lists the tested kinetic reaction mechanisms in this present study in terms of the number of species and reactions. The experimental measurements for data validation were selected from the published work of [19]. The authors employed a stagnation flame configuration system to determine the mole fraction of  $\text{N}_2\text{O}$  from the combustion of a 70/30 (vol%)  $\text{NH}_3/\text{H}_2$  blended fuel. This approach was chosen to avoid stabilization issues, which leads to more consistent production of combustion products and enables the obtaining of accurate measurements.

To examine the effect of equivalence ratio on the  $\text{N}_2\text{O}$  mole fractions, various equivalence ratios have been applied in the range of 0.57–1.4. A top stagnation plate was fixed 2 cm above the outlet section of the burner to manage generating a stagnation flow. The values of the top plate surface temperature ( $T_w$ ) and the mixture inlet velocity ( $V_{in}$ ) were varied because of the variation in equivalence ratios and this variation changed the laminar burning velocity. The experimental data from [19] have been selected as it matched the conditions of our interest in terms of  $\text{NH}_3/\text{H}_2$  ratio, the range of the equivalence ratio and the standard

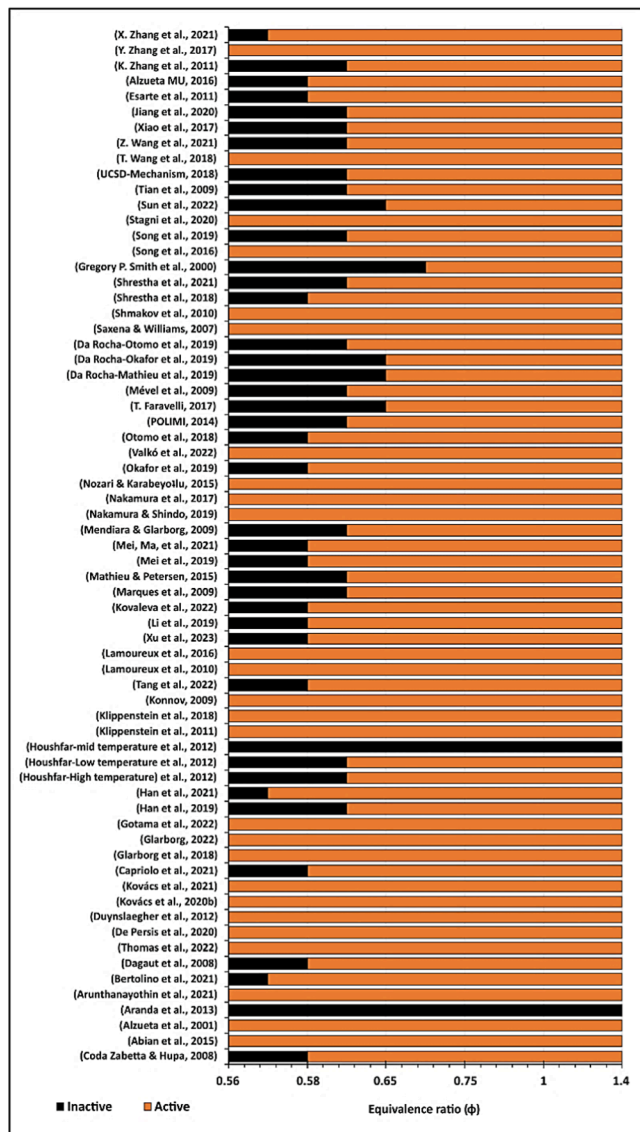


Fig. 5. Comparative analysis of kinetic reaction mechanisms in terms of evaluating activation performance for ignition versus inactivity resulting in ignition failure.

conditions of the unburned gas. In the current study, a normalized error method was chosen as an initial metric for adjusting the discrepancy between a numerical result and an experimental finding [19], considering the uncertainty in the measurements, as illustrated in Eq.1. The  $N_2O$  mole fractions obtained from experimental data tend to be near zero, especially in the  $\phi$  range of 0.65–1.4, which results in considerable uncertainties. Therefore, adjusting for uncertainty is important as it provides context for the discrepancy. A minor difference between the experimental and theoretical values could be meaningful if the uncertainty is very low, while a more substantial difference might be less relevant if the uncertainty is high.

$$\text{Normalised Error} = \frac{F_t - A_t}{\text{Uncertainty}} \quad (1)$$

Where  $F_t$  is the forecast from numerical calculations; and  $A_t$  is the actual value from experiments.

### 3. Results and discussion

Fig. 1 and Fig. 2 illustrate the performance of the mechanisms by

comparing their simulation results with experimental measurements, taking into account the uncertainty in these measurements within the ranges of  $\phi$  (0.57–0.65) and (1–1.4), respectively. This approach helps identify reaction mechanisms that demonstrate a low discrepancy with the experimental results. Meanwhile, Fig. 3 displays the variation in the mole fraction of  $N_2O$  as a function of the equivalence ratio over the range of 0.56 to 1.4. The mole fraction of  $N_2O$  was measured at the end of the computational domain (at  $X = 2$  cm), corresponding to the sampling point for the experimental emissions under steady-state conditions. As shown in Fig. 3, and in line with the experimental measurements reported by the Hayakawa et al. [19], the  $N_2O$  concentration drops exponentially in the range of  $\phi = 0.56$ –0.7, to very low undefined values (the authors consistently reported negative concentration values, with a mean of  $-1.92$  ppm and a standard deviation of  $0.43$  ppm, in the  $\phi = 0.7$ –0.9 range). Subsequently, the concentration increases, reaching a peak of over  $7.5$  ppm at the stoichiometric condition, and then decreases to  $2$  ppm as it is approaching the rich  $\phi = 1.4$  condition. These systematic changes at low concentrations are captured by the selected mechanisms, as depicted in Fig. 3. The one standard deviation of statistical noise for the  $N_2O$  data in the  $\phi = 1.0$ –1.4 region, estimated using the Minimal Spline Fit code [81], was found to be  $0.4$  ppm. This value is 50–150 times lower than the error bars proposed by the authors, suggesting that the evaluated uncertainty in this range is primarily composed of systematic errors and might be overly conservative.

Regarding the uncertainty of equivalence ratios, in the study by Hayakawa [19], the authors carefully analysed the  $N_2O$  concentration uncertainties. However, they did not discuss the uncertainty of equivalence ratios and their impact. Instead, they noted in their paper that the accuracy of the mass flow meter for  $NH_3$  and  $H_2$  is  $\pm 1.0$  % FS, and for air, it is  $\pm 1.5$  % FS. Considering the information provided, the Gaussian error propagation results in a  $1.8$  % uncertainty in equivalence ratios, corresponding to a  $0.01$  uncertainty at the leanest conditions. They also applied parallel measurements at rounded equivalence ratios to assess the reproducibility of the measurements used in their investigations. The scatter of data from parallel measurements is particularly noticeable at lean equivalence ratios (0.57 and 0.585), which should be attributed to the significant value of equivalence ratio uncertainty (approximately  $0.01$ ). For assessing mechanism performance under these conditions, the average of the measured values was taken, with an increased error bar derived from the deviation between the two values ( $\Delta = x_1 - x_2$ ), and their uncertainties (error bars correspond to  $\pm 2\sigma_{1,2}$ ). The uncertainties are calculated as follows:  $\sigma_{\text{mean}}^2 = (\Delta/2)^2 + (\sigma_1^2 + \sigma_2^2)/2$ .

Additional validation using jet-stirred reactor measurements has been adopted to assess the accuracy and generalisability of the well-performing kinetic models in the burner-stabilised stagnation configuration. This approach, based on the study by Zhang et al. [18], involves 70/30 vol%  $NH_3/H_2$  mixtures at atmospheric pressure, covering temperatures from 800 to 1280 K and two equivalence ratios,  $\phi$  (0.25 and 1). The simulation process utilised a transient, perfectly stirred reactor model. Simulation parameters such as end times, boundary conditions, and mesh conditions were carefully applied, following the methodologies detailed in the study by Zhang et al. [18].

As illustrated in Fig. 4, the Glarborg kinetic model exhibits excellent performance in the temperature range of 900 K to 1120 K when  $\phi$  is set to 0.25. However, its performance declines above 1150 K, marked by an overprediction of the  $N_2O$  mole fraction. In contrast, the Klippenstein model consistently overpredicts across the tested temperature range. When the equivalence ratio is increased to 1, the Glarborg model maintains its performance, showing a similar prediction trend as at a  $\phi$  of 0.25. Meanwhile, the Klippenstein model shows improved performance, accurately predicting within the range of 900 K to 1040 K but overestimating  $N_2O$  mole fractions above 1040 K. The kinetic model of Zhang et al. [18], refined based on the provided experimental measurements, as shown in Fig. 4, was also used to validate the experimental

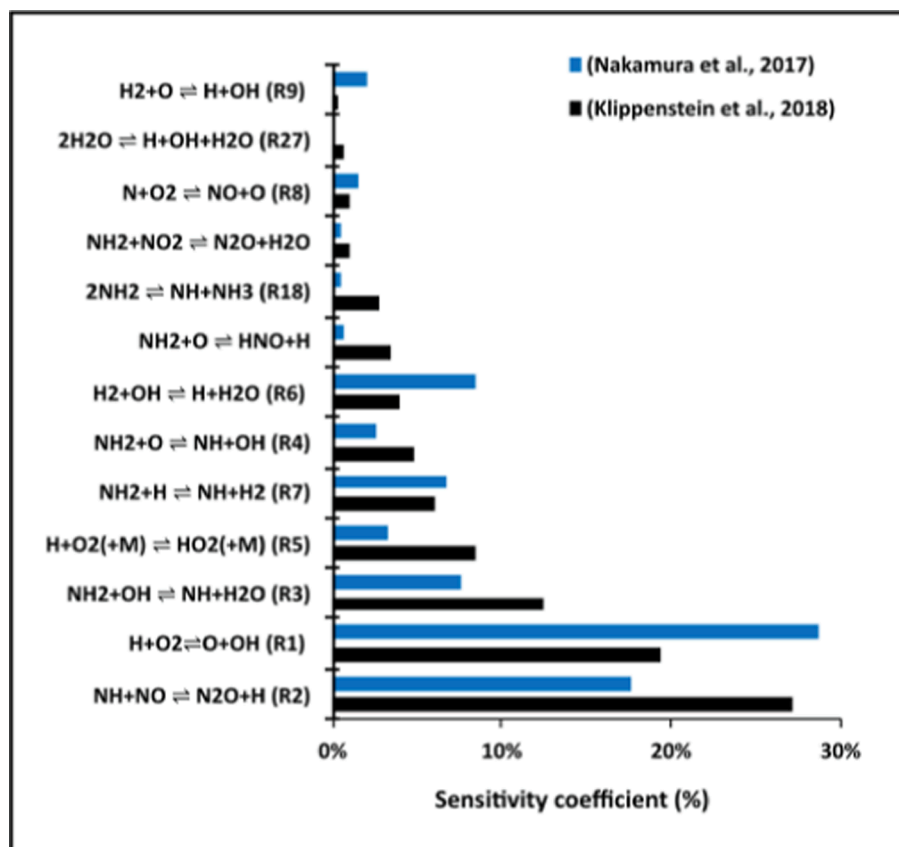


Fig. 6. Reactions with the largest positive sensitivity coefficients for  $\text{N}_2\text{O}$  mole fractions in 70/30 vol%  $\text{NH}_3/\text{H}_2$  premixed flame at  $\phi = 0.6$  in the Klippenstein and Nakamura kinetic models.

measurements of Hayakawa et al. [19], as depicted in Fig. 3. This comparison was conducted to assess its predictive accuracy against other selected models and combustion configurations. The Zhang mechanism provides an excellent estimation of  $\text{N}_2\text{O}$  mole fractions, closely matching the trend estimated by the Glarborg model, except for a tendency to overpredict  $\text{N}_2\text{O}$  mole fractions when  $\phi$  exceeds 0.6. Under these conditions, the Klippenstein model falls within the range of uncertainty measurements, offering a better estimation.

Given the varying trends of  $\text{N}_2\text{O}$  mole fractions in the burner-stabilized stagnation flame model at different equivalence ratios, the analysis has been divided into three categories: lean ( $\phi = 0.6$ ), stoichiometric ( $\phi = 1$ ), and rich flames ( $\phi = 1.4$ ). This categorization is based on the modelling of 70/30 vol%  $\text{NH}_3/\text{H}_2$  blend within this specific flame model. A sensitivity analysis, along with an investigation of the rate of  $\text{N}_2\text{O}$  production/consumption, will be conducted for each category. The aim is to clarify the differences among the selected mechanisms and to identify the key reactions contributing to  $\text{N}_2\text{O}$  formation.

Fig. 1 clearly shows that most of the tested kinetic reaction mechanisms are not included, particularly in the range of 0.57 to 0.6, due to performance issues. The simulations of these models yield zero mole fractions of  $\text{N}_2\text{O}$ , in contrast to the experimental measurements, which indicate a high value of  $\text{N}_2\text{O}$  concentration, as seen in Fig. 3. Further investigation revealed that most of the tested mechanisms failed to meet the combustion requirements under the specified conditions; this was indicated by the ammonia concentration remaining constant, showing no consumption. This occurred despite the solution being grid-independent, as illustrated in Fig. 5.

Consequently, these mechanisms have been excluded from the investigation under their determined conditions, as shown in Fig. 5. The failure to achieve ignition, as observed, might be attributed to kinetic inhibition, a scenario where the reaction rates of key steps in the

mechanism are too slow to initiate or sustain combustion. Additionally, the mechanisms proposed by Houshfar et al. [58] for  $\text{NO}_x$  chemistry in the medium-temperature range, along with those by Aranda et al. [43] have been excluded from further investigations. This exclusion is based on two primary considerations: the absence of a comprehensive sub-mechanism for  $\text{NH}_3$  in Aranda's kinetic model and the lack of  $\text{N}_2\text{O}$  chemistry in Houshfar's chemistry database.

### 3.1. Lean flame conditions

Fig. 3 illustrates the variation in the mole fraction of  $\text{N}_2\text{O}$  as a function of the equivalence ratio ( $\phi$ ) within the range of 0.56 to 1.4. It demonstrates that the  $\text{N}_2\text{O}$  mole fraction decreases sharply, approaching nearly zero, as  $\phi$  increases. According to the normalized error trends in lean flame conditions (0.585–0.65), as shown in Fig. 1, the reaction mechanisms of Glarborg [50] and Klippenstein [56] provide a good estimation of  $\text{N}_2\text{O}$  mole fraction when their outcomes are compared with the experimental measurements, considering the measurement uncertainty along the lean range of 70/30 vol%  $\text{NH}_3/\text{H}_2$  blended fuel. However, their performance deteriorates at an equivalence ratio of 0.57, where the Klippenstein mechanism underestimates the mole fraction of  $\text{N}_2\text{O}$  with a normalized error by 3.4, and the Glarborg mechanism by 2.5. Their performance improves, however, as the equivalence ratio increases, where their normalized error values are less than one. According to Fig. 1, the Nakamura kinetic model [14] shows a high level of accuracy in estimating the error for  $\text{N}_2\text{O}$  mole fractions at  $\phi = 0.6$ , while Klippenstein's mechanism demonstrates better performance than Glarborg's model. Since all the aforementioned mechanisms have minor differences in normalized error, it is worthwhile to investigate the reasons behind this. Therefore, both kinetic models, i.e., Nakamura and Klippenstein, will be investigated in detail with local sensitivity and rate

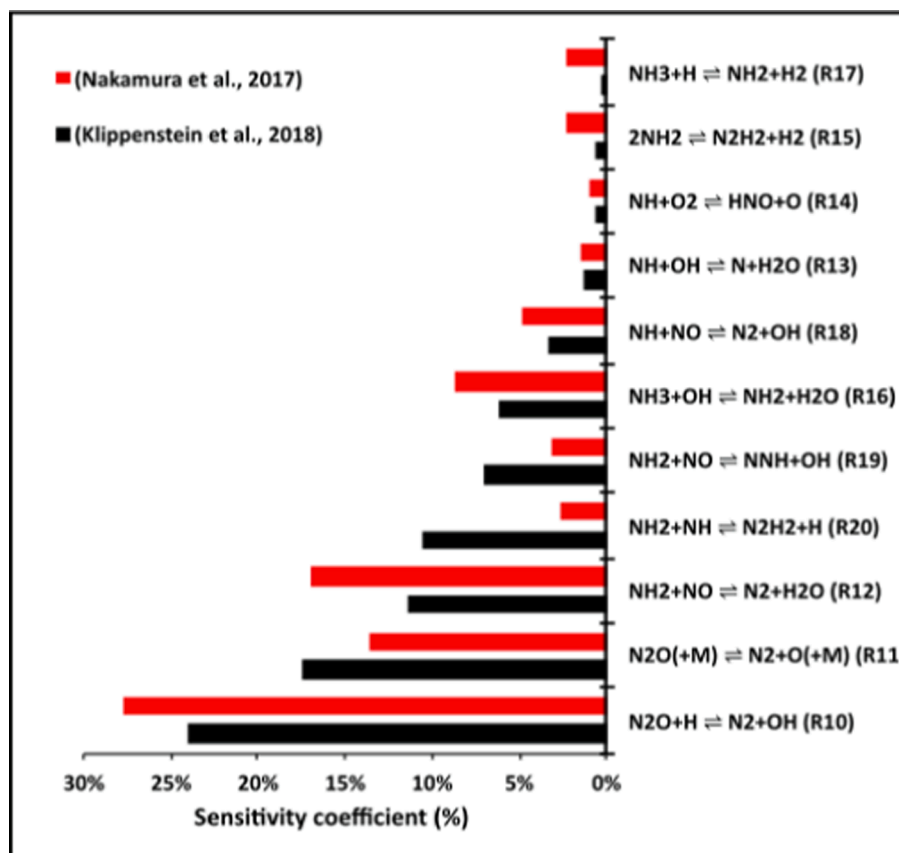


Fig. 7. Reactions with the largest positive sensitivity coefficients for  $\text{N}_2\text{O}$  mole fractions in 70/30 vol%  $\text{NH}_3/\text{H}_2$  premixed flame at  $\varphi = 0.6$  predicted by the Klippenstein and Glarborg kinetic models.

of production/consumption analysis of  $\text{N}_2\text{O}$  to reveal the reasons behind the discrepancies among the selected mechanisms.

Fig. 6 and Fig. 7 show the reactions with the largest positive and negative sensitivity coefficients for  $\text{N}_2\text{O}$  mole fraction in the Nakamura and Klippenstein mechanisms. These coefficients are normalized to their respective sums and presented as percentages. As observed in Fig. 6, reactions  $\text{H} + \text{O}_2 \rightleftharpoons \text{O} + \text{OH}$  (R1) and  $\text{NH} + \text{NO} \rightleftharpoons \text{N}_2\text{O} + \text{H}$  (R2) play a major role in increasing  $\text{N}_2\text{O}$  mole fractions; they also enhance the system reactivity by generating reactive H, O, and OH radicals. Additionally, the promoting effect of the reactions  $\text{NH}_2 + \text{OH} \rightleftharpoons \text{NH} + \text{H}_2\text{O}$  (R3),  $\text{NH}_2 + \text{O} \rightleftharpoons \text{NH} + \text{OH}$  (R4),  $\text{H} + \text{O}_2(+\text{M}) \rightleftharpoons \text{HO}_2(+\text{M})$  (R5),  $\text{H}_2 + \text{OH} \rightleftharpoons \text{H} + \text{H}_2\text{O}$  (R6), and  $\text{NH}_2 + \text{H} \rightleftharpoons \text{NH} + \text{H}_2$  (R7) is clear in both selected mechanisms, contributing to an increase in  $\text{N}_2\text{O}$  mole fraction through the production of more NH and OH radicals. These NH radicals are highly reactive and tend to react with NO to produce  $\text{N}_2\text{O}$  via the reaction  $\text{NH} + \text{NO} \rightleftharpoons \text{N}_2\text{O} + \text{H}$  (R2). Moreover, both selected kinetic models show varying trends in sensitivity coefficients. The reaction steps R1, R6,  $\text{N} + \text{O}_2 \rightleftharpoons \text{NO} + \text{O}$  (R8), and  $\text{H}_2 + \text{O} \rightleftharpoons \text{H} + \text{OH}$  (R9) exhibit higher levels of positive sensitivity coefficients towards promoting the formation of  $\text{N}_2\text{O}$  mole fractions in Nakamura's model compared to the Klippenstein mechanism. This discrepancy is attributed to the mechanistic differences between the tested models, specifically in terms of the rate parameters of the aforementioned reactions, which dictate the reaction rates of each key reaction (see Table 3).

As shown in Fig. 7, both selected kinetic models identify the same key reactions involved in mitigating  $\text{N}_2\text{O}$  formation. The reaction steps  $\text{N}_2\text{O} + \text{H} \rightleftharpoons \text{N}_2 + \text{OH}$  (R10),  $\text{N}_2\text{O} (+\text{M}) \rightleftharpoons \text{N}_2 + \text{O} (+\text{M})$  (R11), and  $\text{NH}_2 + \text{NO} \rightleftharpoons \text{N}_2 + \text{H}_2\text{O}$  (R12) play a significant role in reducing the mole fraction of  $\text{N}_2\text{O}$ , as they exhibit large negative sensitivity coefficients that impact  $\text{N}_2\text{O}$  production. A negative sensitivity coefficient for a reaction implies that an increase in the reaction rate leads to a decrease in

the concentration of the species involved. Therefore, the negative coefficients for reactions R10, R11, and R12 in both selected mechanisms suggest that an increased rate of these reactions results in a lowered concentration of  $\text{N}_2\text{O}$ . Notably, the reaction steps  $\text{NH} + \text{OH} \rightleftharpoons \text{N} + \text{H}_2\text{O}$  (R13) and  $\text{NH} + \text{O}_2 \rightleftharpoons \text{HNO} + \text{O}$  (R14) show roughly the same sensitivity coefficient values in both models. Furthermore, the reaction steps R10, R12,  $2\text{NH}_2 \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}_2$  (R15),  $\text{NH}_3 + \text{OH} \rightleftharpoons \text{NH}_2 + \text{H}_2\text{O}$  (R16),  $\text{NH}_3 + \text{H} \rightleftharpoons \text{NH}_2 + \text{H}_2$  (R17), and  $\text{NH} + \text{NO} \rightleftharpoons \text{N}_2 + \text{OH}$  (R18) show higher negative sensitivity coefficient values in Nakamura's model compared to those predicted by Klippenstein's kinetic model. Conversely, Klippenstein's model demonstrates slightly higher levels of sensitivity coefficients towards the reaction steps R11,  $\text{NH}_2 + \text{NO} \rightleftharpoons \text{NNH} + \text{OH}$  (R19), and  $\text{NH}_2 + \text{NH} \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}$  (R20) than Nakamura's model. The variation in sensitivity coefficients is attributed to the differences in rate parameters between the models, which govern the rate coefficient of each reaction step, as detailed in Table 3.

To explain the reasons behind the discrepancies in the performance of the Klippenstein mechanism [56] compared to the Glarborg kinetic model [50], a sensitivity analysis was conducted under the same conditions ( $\varphi = 0.6$ ) for both models. This analysis aimed to identify which reactions or parameters critically influence the behaviour of the system in each model. Fig. 8 and Fig. 9 showcase the reactions with the largest positive and negative sensitivity coefficients for  $\text{N}_2\text{O}$  mole fraction in the Glarborg and Klippenstein mechanisms, respectively. These Figures clearly illustrate the differences in mechanistic behaviour, particularly in terms of the rate parameters that control the key reactions. Both selected mechanisms highlight the crucial role of reaction steps R1, R2, R3, and R5 in promoting the formation of  $\text{N}_2\text{O}$ . Notably, both models exhibit nearly identical trends for sensitivity coefficients, with the exception of the key reactions R1 and R5, which is predicted by Glarborg's model to have a higher sensitivity coefficient value for

**Table 3**Key reactions of N<sub>2</sub>O formation generated from Klippenstein, Nakamura, Zhang, and Sun kinetic models.

| NO.   | Reaction                                        | (Klippenstein et al., 2018) |        |         | (Nakamura et al., 2017) |        |        | (Y. Zhang et al, 2017) |        |        | (Sun et al, 2022) |        |        |
|-------|-------------------------------------------------|-----------------------------|--------|---------|-------------------------|--------|--------|------------------------|--------|--------|-------------------|--------|--------|
|       |                                                 | A                           | n      | E       | A                       | n      | E      | A                      | n      | E      | A                 | n      | E      |
| (R1)  | $H + O_2 \rightleftharpoons O + OH$             | 1.00E+14                    | 0.00   | 15,286  | 1.040E+14               | 0.00   | 15,286 | 1.000E+14              | 0.00   | 15,286 | 1.00E+14          | 0.00   | 15,286 |
| (R2)  | $NH + NO \rightleftharpoons N_2O + H$           | 2.7E+15                     | -0.780 | 20      | 1.800E+14               | -0.351 | -244   | 1.80E+14               | 0.351  | -244   | 3.65E+14          | -0.45  | 0.00   |
| (R3)  | $NH_2 + OH \rightleftharpoons NH + H_2O$        | 3.3E+06                     | 1.949  | -217    | 4.0E+06                 | 2.00   | 1000   | 3.3E+06                | 1.949  | -217   | 3.3E+06           | 1.949  | -217   |
| (R4)  | $NH_2 + O \rightleftharpoons NH + OH$           | 8.6E-1                      | 4.010  | 1673    | 6.750E+12               | 0      | 0      | —                      | —      | —      | —                 | —      | —      |
| (R5)  | $H + O_2(+M) \rightleftharpoons HO_2(+M)$       | 4.7E+12                     | 0.440  | 0       | 4.650E+12               | 0.44   | 0      | 4.650E+12              | 0.440  | 0      | 2.8E18            | -0.86  | 0.00   |
| (R6)  | $H_2 + OH \rightleftharpoons H + H_2O$          | 2.2E+08                     | 1.51   | 3430    | 4.380E+13               | 0.00   | 6990   | 4.380E+13              | 0.00   | 6990   | 2.16E+08          | 1.51   | 3430   |
| (R7)  | $NH_2 + H \rightleftharpoons NH + H_2$          | 2.1E+13                     | 0.000  | 15,417  | 6.920E+13               | 0.00   | 3650   | 7.2E+05                | 2.32   | 799    | 1.000E+6          | 2.32   | 7990   |
| (R8)  | $N + O_2 \rightleftharpoons NO + O$             | 6.4E+09                     | 1.000  | 6280    | 6.4E+09                 | 1.00   | 6280   | 9.03E+09               | 1.00   | 6498   | 9.0E+09           | 1.00   | 6500   |
| (R9)  | $H_2 + O \rightleftharpoons H + OH$             | 3.8E+12                     | 0.000  | 7948    | —                       | —      | —      | 5.080E+04              | 2.67   | 6292   | 3.87E+4           | 2.7    | 6260   |
| (R10) | $N_2O + H \rightleftharpoons N_2 + OH$          | 6.4E+07                     | 1.835  | 13,492  | 3.310E+10               | 0.00   | 5090   | 6.4E+07                | 1.835  | 13,492 | 3.31E+10          | 0.00   | 5090   |
| (R11) | $N_2O(+M) \rightleftharpoons N_2 + O(+M)$       | 6.0E+14                     | 0.000  | 57,444  | 6.620E+14               | 0.00   | 57,500 | 7.2E+14                | 0.000  | 57,410 | 6.62E+14          | 0.00   | 57,500 |
| (R12) | $NH_2 + NO \rightleftharpoons N_2 + H_2O$       | 2.6E+19                     | -2.369 | 870     | —                       | —      | —      | 1.36E+16               | -1.25  | 0.0    | 3.10E+13          | -0.48  | 1180   |
| (R13) | $NH + OH \rightleftharpoons N + H_2O$           | 1.6E+07                     | 1.733  | -576    | 5.0E+11                 | 0.50   | 2000   | 1.6E+07                | 1.733  | -576   | —                 | —      | —      |
| (R14) | $NH + O_2 \rightleftharpoons HNO + O$           | 2.4E+13                     | 0.00   | 13,850  | 1.0E+13                 | 0.00   | 12,000 | 1.75E+12               | 0.114  | 11,055 | —                 | —      | —      |
| (R15) | $2NH_2 \rightleftharpoons N_2H_2 + H_2$         | 1.7E+08                     | 1.62   | 11,783  | 5.0E+13                 | 0.00   | 10,000 | —                      | —      | —      | —                 | —      | —      |
| (R16) | $NH_3 + OH \rightleftharpoons NH_2 + H_2O$      | 2.0E+06                     | 2.040  | 566     | 3.250E+12               | 0.00   | 2120   | 2.0E+06                | 2.040  | 566    | 2.0E+06           | 2.040  | 566    |
| (R17) | $NH_3 + H \rightleftharpoons NH_2 + H_2$        | 6.4E+05                     | 2.390  | 10,171  | 6.40E+05                | 2.390  | 10,171 | —                      | —      | —      | 6.4E+05           | 2.390  | 10,171 |
| (R18) | $NH + NO \rightleftharpoons N_2 + OH$           | 6.8E+14                     | -0.780 | 20      | —                       | —      | —      | 2.70E+12               | -0.072 | -512   | 2.70E+12          | -0.072 | -512   |
| (R19) | $NH_2 + NO \rightleftharpoons NNH + OH$         | 4.3E+10                     | 0.294  | -866    | 3.100E+13               | -0.48  | 1180   | 3.100E+13              | -0.48  | 1180   | 4.3E10            | 0.294  | -870   |
| (R20) | $NH_2 + NH \rightleftharpoons N_2H_2 + H$       | 4.3E+14                     | -0.272 | -77     | 1.500E+15               | -0.50  | 0.00   | 4.30E+14               | -0.272 | -77    | 1.50E+15          | -0.5   | 0.00   |
| (R21) | $2NH_2 \rightleftharpoons NH + NH_3$            | 5.6E+00                     | 3.530  | 552     | —                       | —      | —      | —                      | —      | —      | 5.6000E0          | 3.53   | 552    |
| (R22) | $NH_2 + O \rightleftharpoons HNO + H$           | 6.6E+13                     | 0.00   | 0.00    | 4.60E+13                | 0.00   | 0.00   | 6.600E+13              | 0.00   | 0.0    | 3.90E+13          | 0.00   | 0.0    |
| (R23) | $NH_2 + NO_2 \rightleftharpoons N_2O + H_2O$    | 2.2E+11                     | 0.110  | -1186   | 1.60E+16                | 1.44   | 268    | 3.0E+14                | 0.77   | 242    | 2.60E+18          | -2.191 | 455    |
| (R24) | $2H_2O \rightleftharpoons H + OH + H_2O$        | 1.0E+26                     | -2.440 | 120,180 | —                       | —      | —      | —                      | —      | —      | —                 | —      | —      |
| (R25) | $N_2O + O \rightleftharpoons 2NO$               | 9.2E+13                     | 0.000  | 27,679  | 9.150E+13               | 0.00   | 27,693 | —                      | —      | —      | —                 | —      | —      |
| (R26) | $N_2O + O \rightleftharpoons N_2 + O_2$         | 9.2E+13                     | 0.000  | 27,679  | 3.690E+12               | 0.00   | 15,944 | —                      | —      | —      | —                 | —      | —      |
| (R27) | $N_2H_2 + NO \rightleftharpoons N_2O + NH_2$    | 4.0E+12                     | 0.000  | 11,922  | 3.000E+10               | 0.00   | 0.00   | 4.0E+12                | 0.0    | 11,922 | 4.00E+12          | 0.00   | 11,922 |
| (R28) | $NH + OH \rightleftharpoons HNO + H$            | 3.2E+14                     | -0.376 | -46     | —                       | —      | —      | 3.20E+14               | -0.376 | -46    | 3.20E+14          | -0.376 | -46.0  |
| (R29) | $N + NO \rightleftharpoons N_2 + O$             | 9.4E+12                     | 0.140  | 0       | 1.00E+14                | 0.00   | 75,490 | 11.28E+12              | 0.14   | 0.0    | 9.40E+12          | 0.14   | 0.00   |
| (R30) | $N_2O + H_2 \rightleftharpoons N_2 + H_2O$      | —                           | —      | —       | —                       | —      | —      | 2.100E+14              | 0.0    | 32,500 | 9.4E+12           | 0.140  | 0      |
| (R31) | $NH + NO_2 \rightleftharpoons N_2O + OH$        | 4.1E+12                     | 0.00   | 0.00    | 1.00E+13                | 0.00   | 0.00   | 4.1E+12                | 0.000  | 0.000  | 4.1E12            | 0.000  | 0.000  |
| (R32) | $2HNO \rightleftharpoons N_2O + H_2O$           | 9.0E+08                     | 0.00   | 3100    | 3.950E+12               | 0.00   | 5000   | 9.0E+08                | 0.00   | 3100   | 3.950E12          | 0.00   | 5000   |
| (R33) | $HNO + NO \rightleftharpoons N_2O + OH$         | 1.20E-4                     | 4.330  | 25,080  | 2.000E+12               | 0.00   | 26,000 | 1.20E-4                | 4.330  | 25,080 | 2.00E+12          | 0.00   | 2600   |
| (R34) | $NNH + O \rightleftharpoons N_2O + H$           | 1.9E+14                     | -0.274 | -22     | —                       | —      | —      | 1.9E+14                | -0.274 | -22    | 1.90E+14          | -0.274 | -22.0  |
| (R35) | $N + OH \rightleftharpoons NO + H$              | 3.8E+13                     | 0.00   | 0.00    | —                       | —      | —      | —                      | —      | —      | —                 | —      | —      |
| (R36) | $H + O_2 + H_2O \rightleftharpoons HO_2 + H_2O$ | —                           | —      | —       | —                       | —      | —      | —                      | —      | —      | 1.126E+19         | -0.76  | 0.00   |
| (R37) | $N_2H_3 \rightleftharpoons N_2H_2 + H$          | 1.3E+14                     | 0.00   | 3871    | —                       | —      | —      | —                      | —      | —      | 3.6E+47           | -10.38 | 69,009 |

\*Units are cm<sup>3</sup>, mol, s, cal, K.

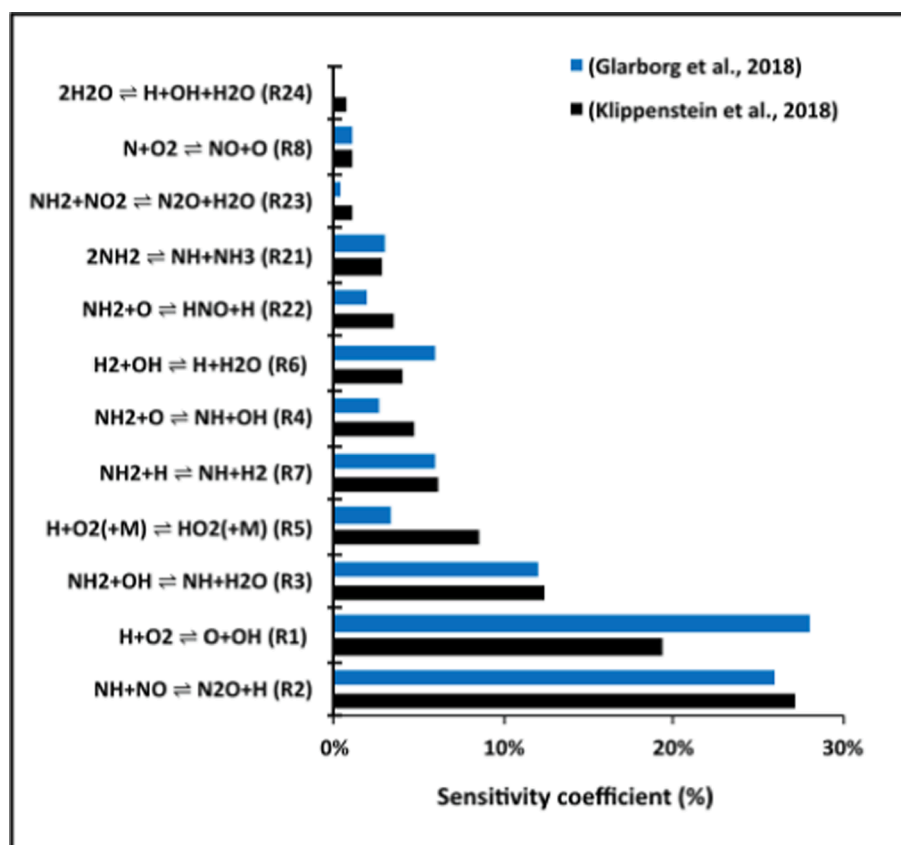


Fig. 8. Reactions with the largest positive sensitivity coefficients for N<sub>2</sub>O mole fractions in 70/30 vol% NH<sub>3</sub>/H<sub>2</sub> premixed flame at  $\phi = 0.6$  in the Klippenstein and Glarborg kinetic models.

promoting the reactivity of the system with O and OH radicals. Similarly, reaction steps R6 and  $2\text{NH}_2 \rightleftharpoons \text{NH} + \text{NH}_3$  (R21) are also shown to have a high sensitivity coefficient value towards N<sub>2</sub>O formation in Glarborg's model. In contrast, Klippenstein's model identifies key reactions R4, R5,  $\text{NH}_2 + \text{O} \rightleftharpoons \text{HNO} + \text{H}$  (R22), and  $\text{NH}_2 + \text{NO}_2 \rightleftharpoons \text{N}_2\text{O} + \text{H}_2\text{O}$  (R23) as having high values of positive sensitivity values, which suggests they are more effective in promoting N<sub>2</sub>O formation compared to Glarborg's model. Additionally, Klippenstein's mechanism highlights the role of the key reaction  $2\text{H}_2\text{O} \rightleftharpoons \text{H} + \text{OH} + \text{H}_2\text{O}$  (R24) in promoting N<sub>2</sub>O formation, whereas this reaction's contribution is notably not observed in Glarborg's mechanism.

The chemistry of N<sub>2</sub>O, for the key reactions that exhibit negative sensitivity coefficients towards N<sub>2</sub>O mole fraction, is consistent across the selected kinetic models by Glarborg and Klippenstein, as illustrated in Fig. 9. Both models identify reaction steps R10, R11, R12, R16, R19 and R20 as having significant negative sensitivity coefficients, indicating their role in reducing the N<sub>2</sub>O mole fraction. However, the trends of these reactions differ, highlighting mechanistic variations between the kinetic models in terms of rate parameters. Glarborg's model demonstrates a pronounced trend for the reaction steps R10, R11, and R12, in contrast to Klippenstein's model, which predicts a higher sensitivity coefficient for reaction step R19 than that in the Glarborg model. This further indicates differences in how each model approaches the N<sub>2</sub>O consumption process.

The discrepancies in predictions among the kinetic models under consideration arise from mechanistic differences. A detailed comparative analysis of the models developed by Klippenstein and Glarborg reveals significant differences, particularly regarding rate parameter values and the definition of reaction steps. The rate parameters, crucial for the predictive accuracy of kinetic models, show notable variances between the two models. Specifically, the Klippenstein model utilizes

distinct rate parameters for reaction steps identified in Table 4, such as  $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{NH}_3 + \text{O}_2$ ,  $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{H}_2\text{NO} + \text{OH}$ ,  $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{H}_2\text{O} + \text{HNO}$ , and  $\text{N} + \text{O}_2 \rightleftharpoons \text{NO} + \text{O}$ , which significantly differ from those employed in the Glarborg model. These differences in rate parameters critically impact the model's ability to accurately simulate reaction kinetics and predict results.

The study by Klippenstein and Glarborg et al. [82] highlights the importance of the  $\text{NH}_2 + \text{HO}_2$  reaction in the ignition and oxidation of NH<sub>3</sub> at high pressures and intermediate temperatures, emphasizing its product channels ( $\rightarrow \text{NH}_3 + \text{O}_2$ ,  $\rightarrow \text{HNO} + \text{H}_2\text{O}$ ,  $\rightarrow \text{H}_2\text{NO} + \text{OH}$ , and  $\rightarrow \text{NH}_2\text{OOH}$ ). The pivotal role of the  $\text{HO}_2 + \text{NH}_2$  reaction has also been supported by studies [63,83,84], indicating that calculations of ignition and oxidation at high pressures and intermediate temperatures are sensitive to the rate constant and branching ratio for  $\text{HO}_2 + \text{NH}_2$ . Moreover, the Klippenstein model outlines a reaction pathway,  $\text{H}_2\text{NO} + \text{HO}_2 \rightleftharpoons \text{H}_2\text{O}_2 + \text{HNO}$ , potentially involving unique intermediate species that are not directly included in the Glarborg model. Due to its significance in the hydrogen abstraction from H<sub>2</sub>NO, this particular reaction plays a pivotal role in the combustion of ammonia, notably at lower temperatures [85].

These mechanistic differences, covering both rate parameter values and specific reaction steps reflect the discrepancies among the kinetic models in terms of laminar flame velocity and speciation mole fractions. The selection of rate parameters and the description of reaction pathways are instrumental in defining the model's predictive capabilities, emphasizing the need for thorough validation against experimental data. Consequently, the observed differences between the Klippenstein and Glarborg models not only illustrate the inherent challenges in kinetic modelling but also underscore the necessity for continuous refinement and experimental comparison to enhance model accuracy and reliability.

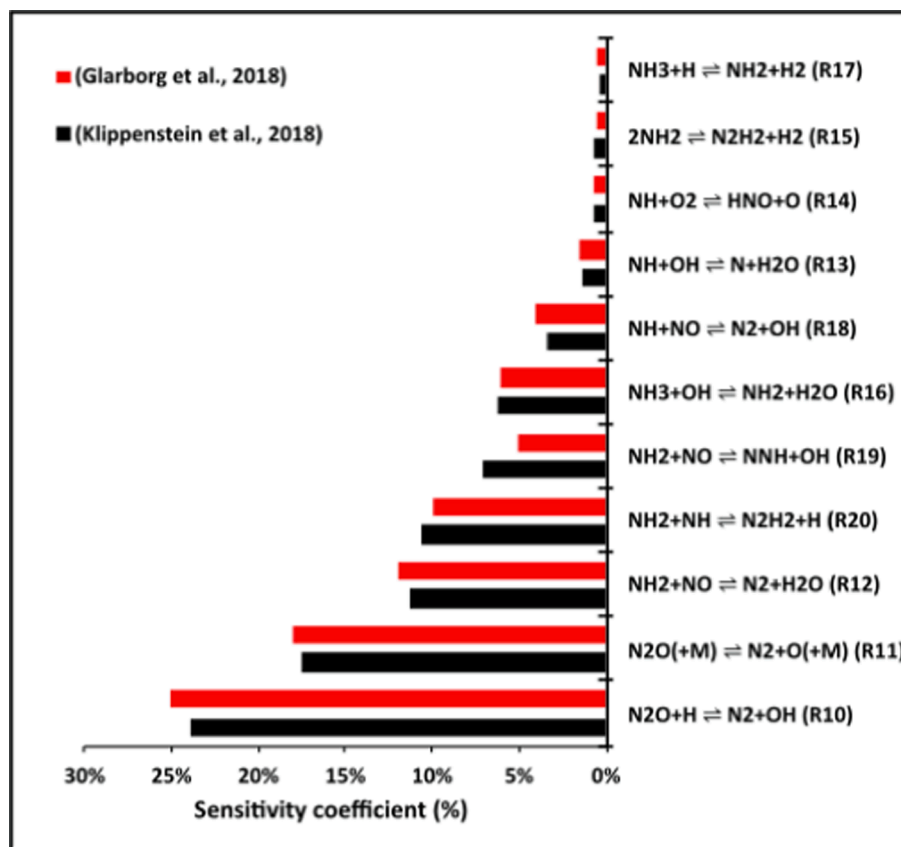


Fig. 9. Reactions with the largest negative sensitivity coefficients for N<sub>2</sub>O mole fractions in 70/30 vol% NH<sub>3</sub>/H<sub>2</sub> premixed flame at  $\phi = 0.6$  in the Klippenstein and Nakamura kinetic models.

**Table 4**  
Key Reactions Illustrating Model Differences Between Klippenstein and Glarborg Kinetic Models.

| Reactions                                                                                  | (Klippenstein et al., 2018) |       |       | (Glarborg et al., 2018) |   |      |
|--------------------------------------------------------------------------------------------|-----------------------------|-------|-------|-------------------------|---|------|
|                                                                                            | A                           | n     | E     | A                       | n | E    |
| $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{NH}_3 + \text{O}_2$                    | 1.70E+04                    | 1.55  | 2027  | 1.5E+14                 | 0 | 0    |
| $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{H}_2\text{NO} + \text{OH}$             | 2.50E+17                    | -1.28 | 1166  | 2.5E+13                 | 0 | 0    |
| $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{H}_2\text{O} + \text{HNO}$ (DUPLICATE) | 1.60E+07                    | 0.55  | 525   | 2.5E+12                 | 0 | 0    |
| $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{H}_2\text{O} + \text{HNO}$ (DUPLICATE) | 5.70E+15                    | -1.12 | 707   | –                       | – | –    |
| $\text{HO}_2 + \text{NH}_2 \rightleftharpoons \text{H}_2\text{O} + \text{HON}$ (DUPLICATE) | 2.10E+07                    | 0.64  | 811   | –                       | – | –    |
| $\text{N} + \text{O}_2 \rightleftharpoons \text{NO} + \text{O}$                            | 6.40E+09                    | 1.00  | 6280  | 5.9E+09                 | 1 | 6280 |
| $\text{H}_2\text{NO} + \text{HO}_2 \rightleftharpoons \text{H}_2\text{O}_2 + \text{HNO}$   | 2.90E+04                    | 2.69  | -1600 | Missing                 |   |      |

\*Units are cm<sup>3</sup>, mol, s, cal, K.

Fig. 10 and Fig. 11 show the variation of N<sub>2</sub>O production and consumption rates as a function of distance, respectively, by the most important N<sub>2</sub>O reactions in the Nakamura and the Klippenstein mechanisms. As can be noticed from Fig. 10 and Fig. 11, the net production rate of N<sub>2</sub>O increases when the temperature of the system increases sharply and then this is followed by a sudden decrease downstream. According to Nakamura's kinetic model, the climbing influence of the total component of N<sub>2</sub>O is governed by the action of reaction R2 which is responsible for 99 % of the N<sub>2</sub>O formation in the combustion zone (see Fig. 12), while the decrease in N<sub>2</sub>O is basically due to the retarding influence of reaction R10 which accounts for 85 % of the summed reaction rate of all N<sub>2</sub>O consuming reactions (see Fig. 13). Similarly,

Klippenstein's kinetic model shows an increase in the total N<sub>2</sub>O due to the R2 reaction and then consumed by reactions R10 and R11, which thus are considered substantial in the consumption of N<sub>2</sub>O (see Fig. 13).

The chemical pathways presented in Fig. 14 also show the dominant role of reaction R2 in the formation of N<sub>2</sub>O, which indicates the dominant role of NH radicals in the production of N<sub>2</sub>O (i.e. accounts for almost 98 %), as well as the substantial part of reaction R10 in the consumption of N<sub>2</sub>O, which leads to production of N<sub>2</sub> and OH.

Fig. 13 shows different trends among the chosen mechanisms. The Nakamura kinetic model shows a higher level of N<sub>2</sub>O consumption by reaction R10 when compared to the Klippenstein model. Conversely, the Klippenstein model shows increased consumption rates for reactions R11, N<sub>2</sub>O + O ⇌ 2NO (R25), N<sub>2</sub>O + O ⇌ N<sub>2</sub> + O<sub>2</sub> (R26), and N<sub>2</sub>H<sub>2</sub> + NO ⇌ N<sub>2</sub>O + NH<sub>2</sub> (R27) relative to those in the Nakamura model. Notably, reaction R27 has a negligible impact within the Nakamura mechanism. These differences in reaction sensitivities can be attributed to the variations in the rate parameters that control the reactions within each mechanism (see Table 3), which affect the rate of the selected reaction (see Fig. 15). As noticed from Fig. 15 the Heat Release Rate (HRR) predicted by Klippenstein is greater than that estimated by Nakamura's model.

Fig. 10 and Fig. 11 also depict some insights that are quite significant. According to the study by Alnasif et al. [86], the laminar flame speed for the Klippenstein model (8.66 cm/s) is higher than that predicted by the Nakamura model (6.72 cm/s). This aligns with the well-known behaviour of stagnation flames, where the flame location is closely related to the flame speed. Faster flames usually stabilize closer to the inlet in areas with higher gas velocity. This is clearly shown in Fig. 10 and Fig. 11, where the flame, according to the Klippenstein model, is seen closer to the burner nozzle, while the Nakamura model, which has a slower flame speed, shows the flame stabilizing further from

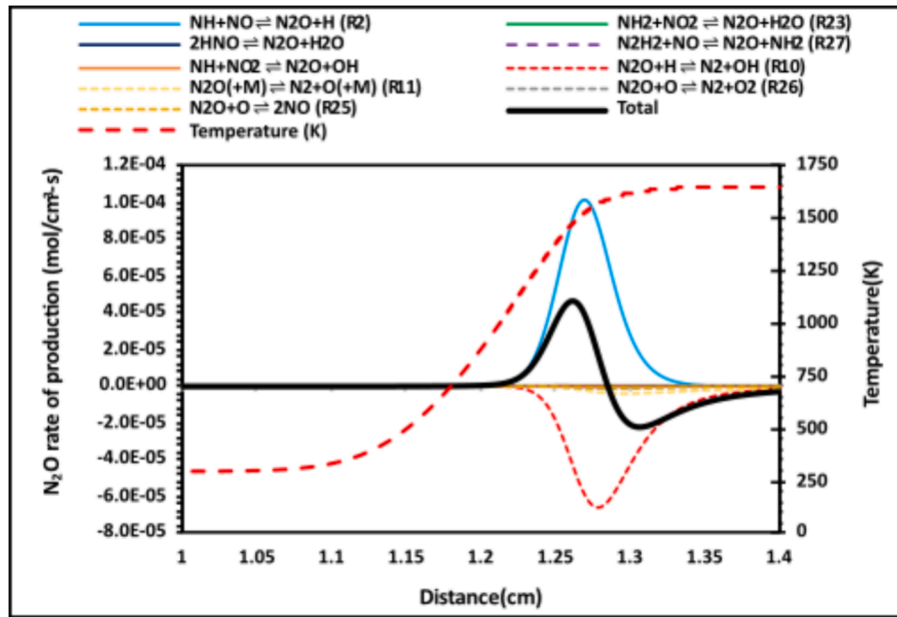


Fig. 10. The rate of production/consumption of  $N_2O$  for 70/30 vol%  $NH_3/H_2$  mixture at 0.6 of  $\phi$  estimated by the Nakamura mechanism.

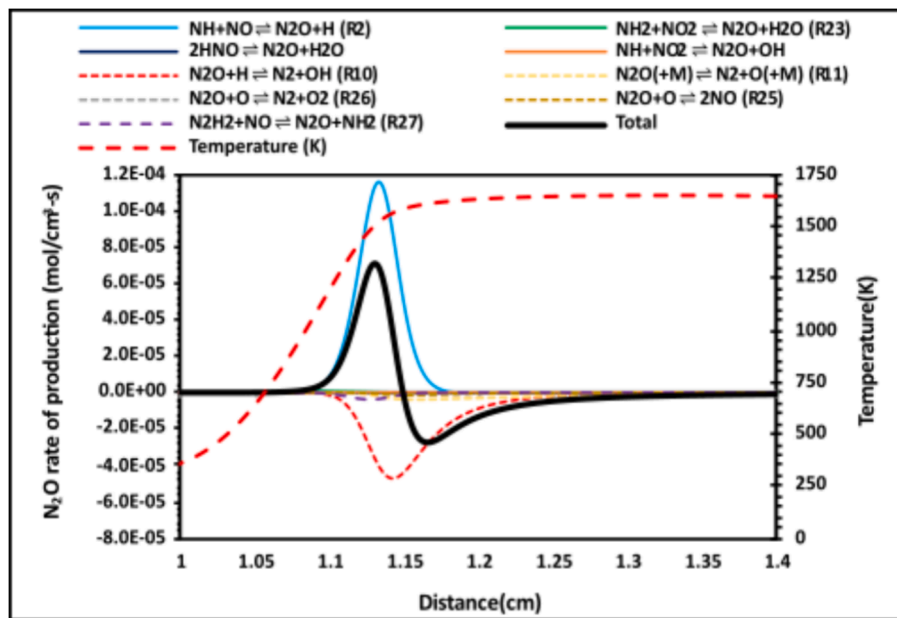


Fig. 11. The rate of production/consumption of  $N_2O$  for 70/30 vol%  $NH_3/H_2$  mixture at 0.6 of  $\phi$  estimated by Klippenstein kinetic model.

the nozzle.

The consumption behaviour of reaction R27 can be explained by the negative trend in the reaction rate of the aforementioned reaction, as shown in Fig. 15. Chemical reactions can shift their equilibrium position based on changes in temperature or the concentrations of reactants and products. A change in temperature, the depletion of a reactant, and the high concentration of products create conditions favourable for the dominance of the reverse reaction ( $N_2O + NH_2 \rightarrow N_2H_2 + NO$ ), where the products react to form the reactants, resulting in an increase in reactant concentrations. This behaviour is clearly observable at 0 cm, as depicted in Fig. 16, where the mole fraction of  $N_2H_2$  gradually increases while the concentration of  $NH_2$  decreases.

The intensity rate of the reaction R27 increases when moving downstream, Fig. 15. According to the Klippenstein model, the reaction step R27 shows a pronounced temperature dependency (refer to

Table 3). The rate of this reaction increases significantly with an increase in temperature, resulting in a marked decrease in the  $NH_2$  mole fraction, as evidenced in Fig. 16, through the reverse reaction ( $N_2O + NH_2 \rightarrow N_2H_2 + NO$ ). As the reaction progresses, reactants, particularly  $NH_2$ , are consumed, resulting in a deceleration of the reaction. As R27 demonstrates a minor impact on the depletion of  $N_2O$ , its influence appears to be potentially overstated in the Klippenstein model. The recent study by Cornell et al. [87] found that the original rate constant of R27 overestimates the mole fraction of  $N_2O$ , suggesting that this reaction is either too slow or has no significant impact on the  $N_2O$  mole fraction. Furthermore, the study observed that models with and without the inclusion of R27 showed nearly identical predictions. This indicates that the experimental datasets used for validation may not have adequately tested this specific reaction (see Section 3.4).

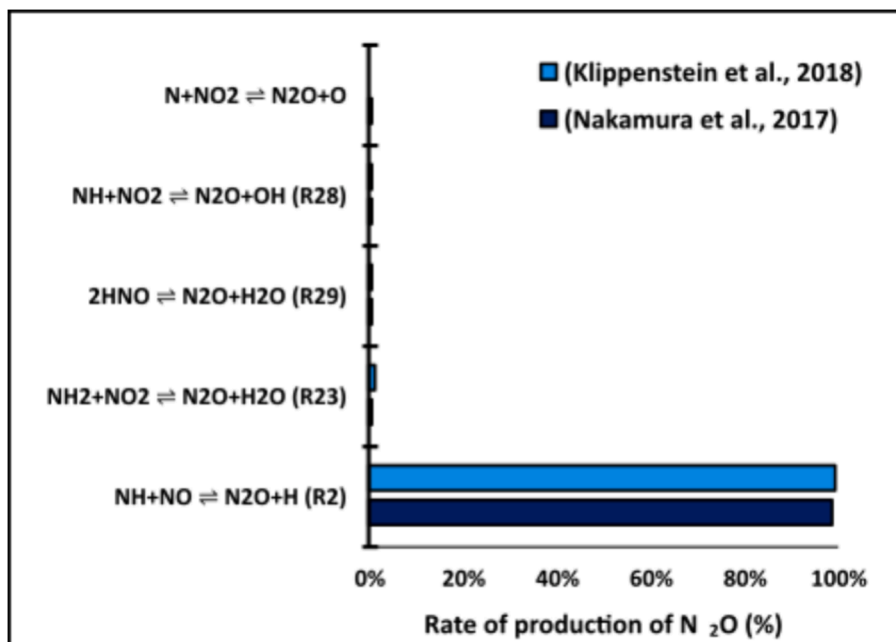


Fig. 12. Rate of production (in %) at  $\phi = 0.6$  estimated by the Nakamura and Klippenstein kinetic models.

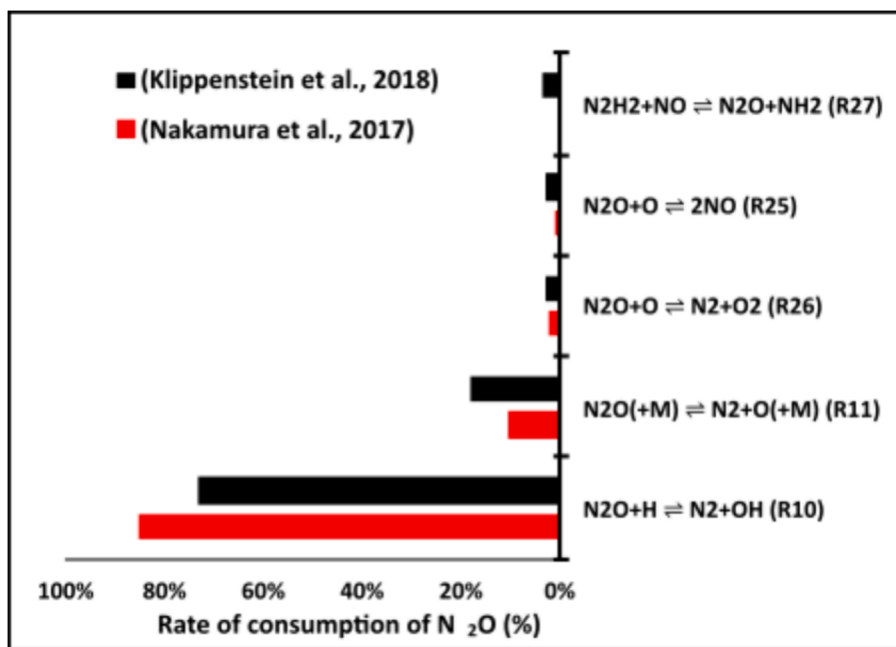


Fig. 13. Rate of consumption (in %) at  $\phi = 0.6$  estimated by the Nakamura and Klippenstein kinetic models.

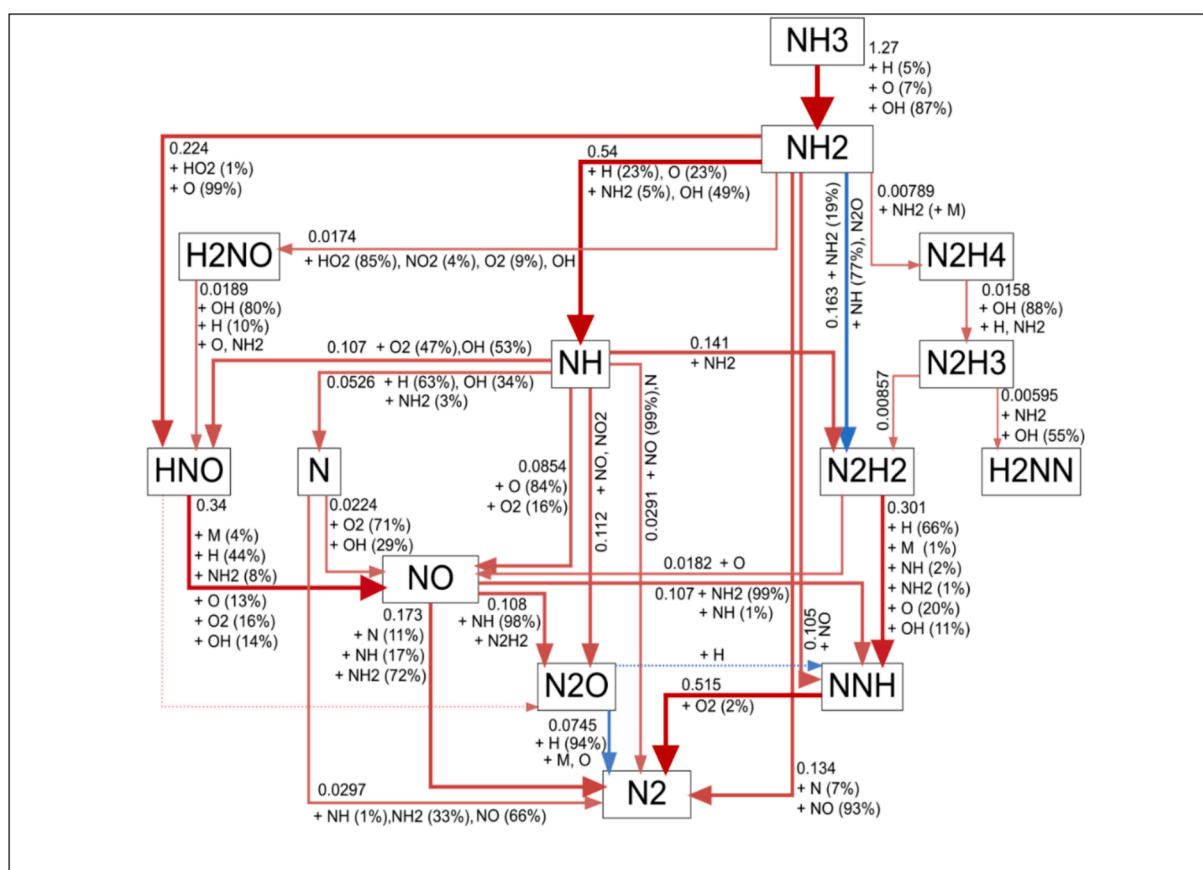
### 3.2. Stoichiometric flame conditions

Based on Fig. 3, it has been observed that the mole fraction of  $N_2O$  has slightly increased. According to the normalised error results, the prediction accuracy of most tested mechanisms falls within the range of experimental uncertainty (Normalised error < 1). This indicates that a larger difference between the experimental and theoretical values is not significant, given the large uncertainty. Particularly at stoichiometric conditions for  $N_2O$  mole fraction, the magnitude of uncertainty exceeds the measured values, rendering the differences between simulations and experiments statistically insignificant and within the expected variability of the measurements. Consequently, the majority of the tested kinetic models demonstrate good performance in predicting  $N_2O$  mole

fractions, as shown in Fig. 2.

Considering the uncertainty of the measurements, the estimation accuracy of the Klippenstein kinetic model falls within the range of the experimental value (normalised error less than 1), as depicted in Fig. 2. Furthermore, as observed from Fig. 2, the Zhang kinetic model [59] exhibits better performance, as its normalised error is lower than that of the Klippenstein model. Therefore, both the Klippenstein and Zhang reaction mechanisms will be analysed in terms of their behaviour in estimating  $N_2O$ , and the reasons behind their discrepancies in this condition will be explored.

Fig. 17 and Fig. 18 show the most influential reactions on  $N_2O$  mole fractions with positive and the negative sensitivity coefficients estimated by Zhang and Klippenstein kinetic mechanisms, respectively. As



which is also visualized by line thickness.

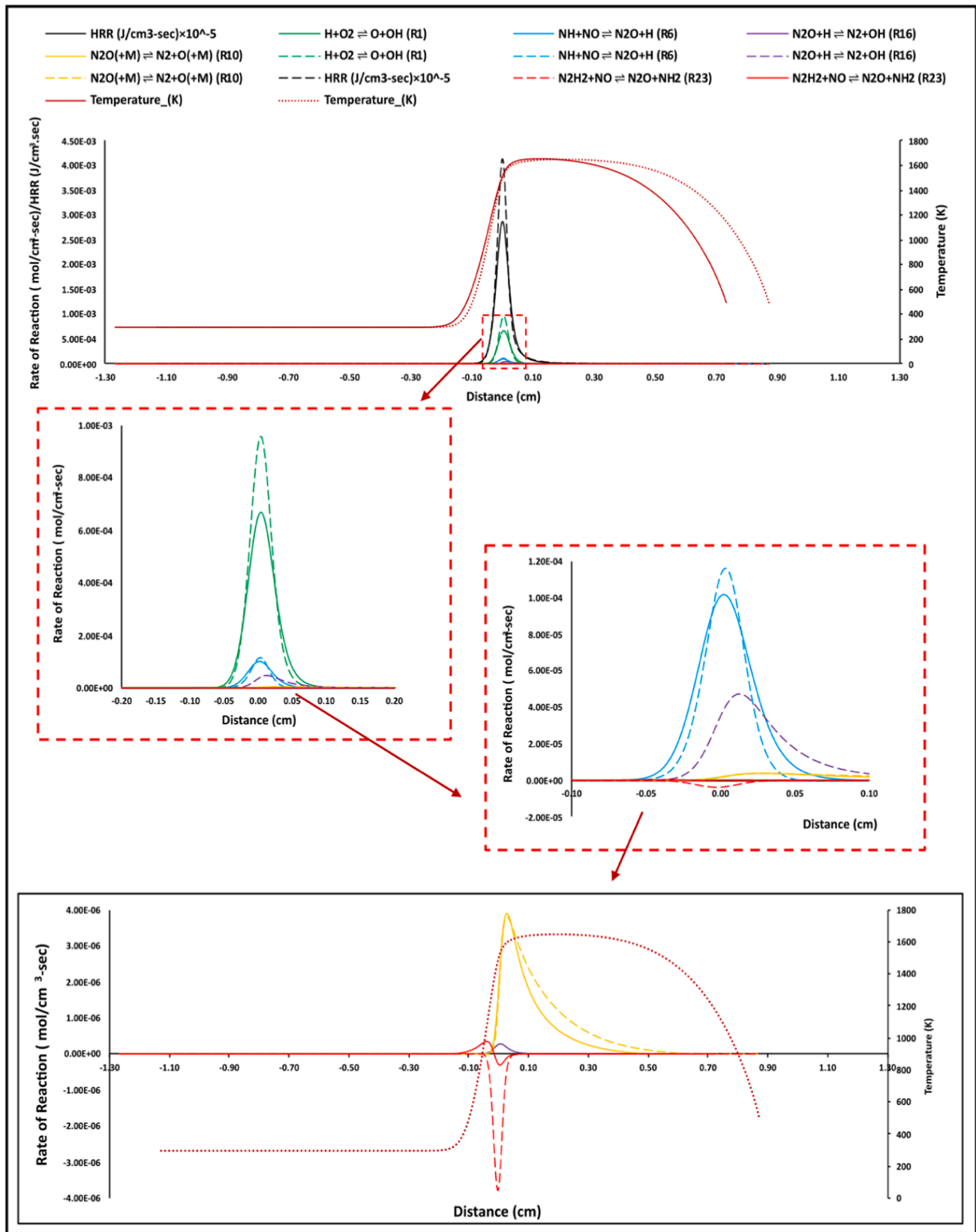


Fig. 15. The reaction rate profiles of reactions most influential to the formation/reduction of  $\text{N}_2\text{O}$  mole fractions for 70/30 vol%  $\text{NH}_3/\text{H}_2$  mixture at  $\phi = 0.6$ . The result for the Nakamura and Klippenstein kinetic models is shown with solid and dashed lines, respectively.

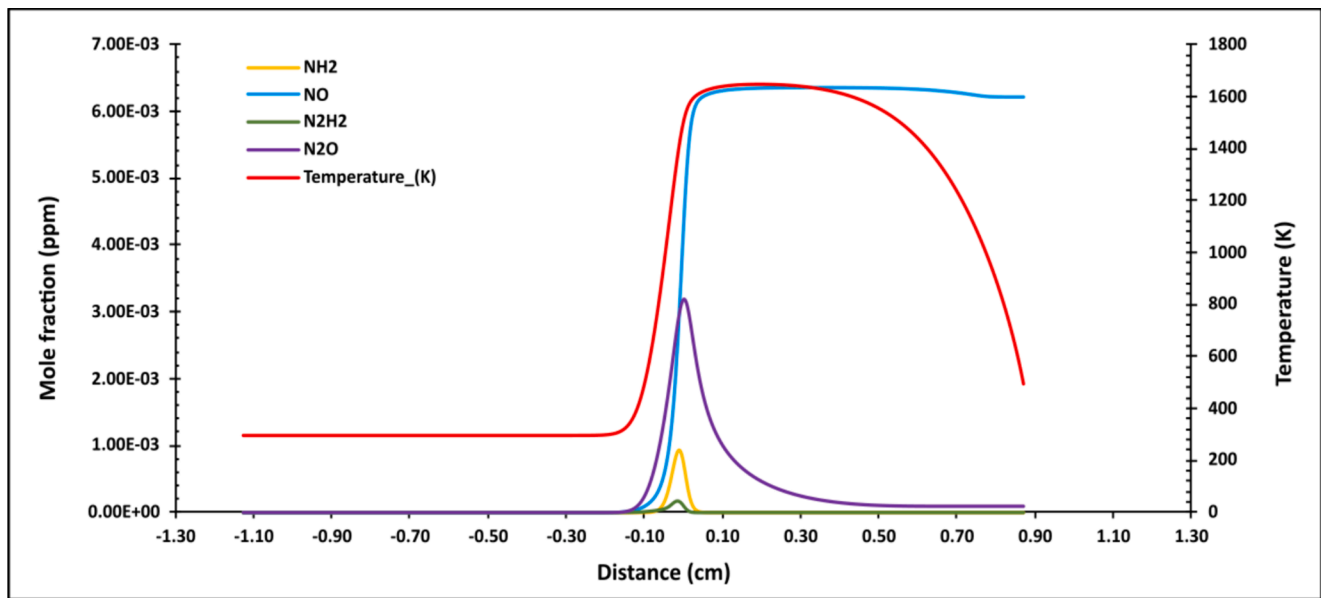


Fig. 16. Variation of  $\text{NH}_2$ ,  $\text{NO}$ ,  $\text{N}_2\text{H}_2$ , and  $\text{N}_2\text{O}$  mole fractions, as well as temperature, as a function of axial distance in a 70/30 vol%  $\text{NH}_3/\text{H}_2$  mixture at  $\phi = 0.6$ , as predicted by the Klippenstein kinetic model.

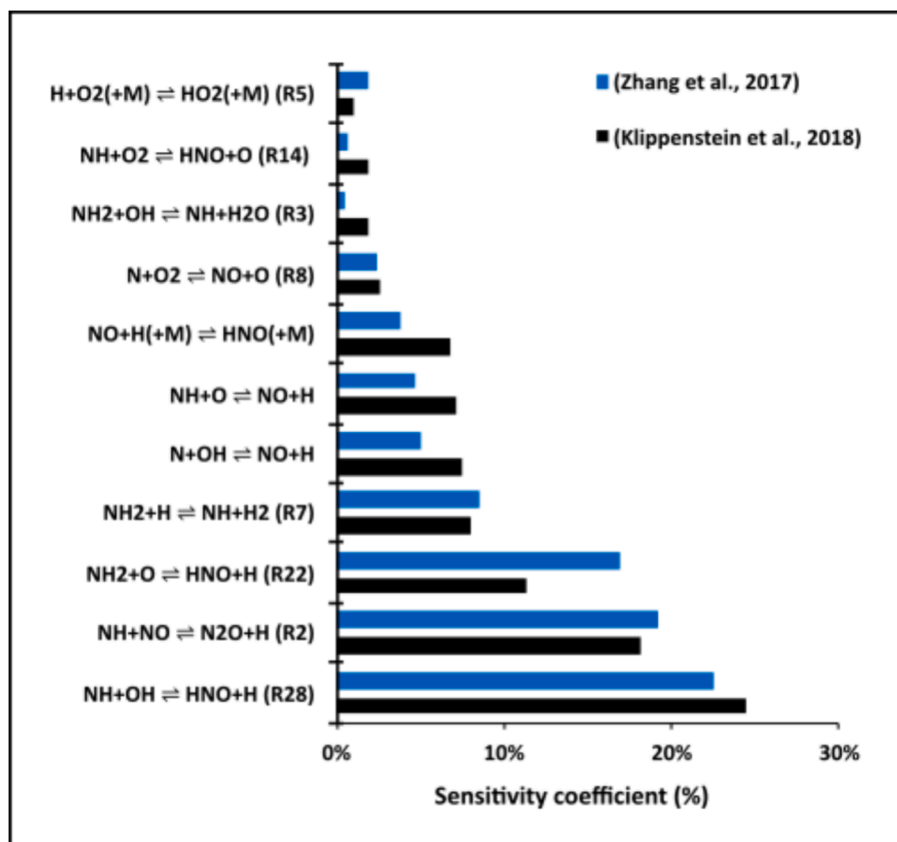


Fig. 17. Reactions with the largest positive local sensitivity coefficients for  $\text{N}_2\text{O}$  mole fractions in 70/30  $\text{NH}_3/\text{H}_2$  vol% premixed flame at  $\phi = 1$  in the Klippenstein and Zhang kinetic models.

shown in Fig. 17, both kinetic models indicate the same reaction steps having a positive impact on  $\text{N}_2\text{O}$  formation, though with different trends in sensitivity values. In both models, the reactions  $\text{NH} + \text{OH} \rightleftharpoons \text{HNO} + \text{H}$  (R28) R2, R7, and R22 are identified as the most dominant in increasing the mole fraction of  $\text{N}_2\text{O}$  by enhancing the H and  $\text{H}_2$  pools within the system, thus improving system reactivity. Klippenstein's model displays

a higher sensitivity trend for the reaction R28 compared to the Zhang model, which, conversely, shows higher sensitivity values for reactions R2, R7, and R22 for  $\text{N}_2\text{O}$  formation. The variation in their sensitivity values is due to the different values of their rate parameters among the selected models, which in turn control their reaction rates. According to Fig. 18, both models indicate the highest negative sensitivity coefficient

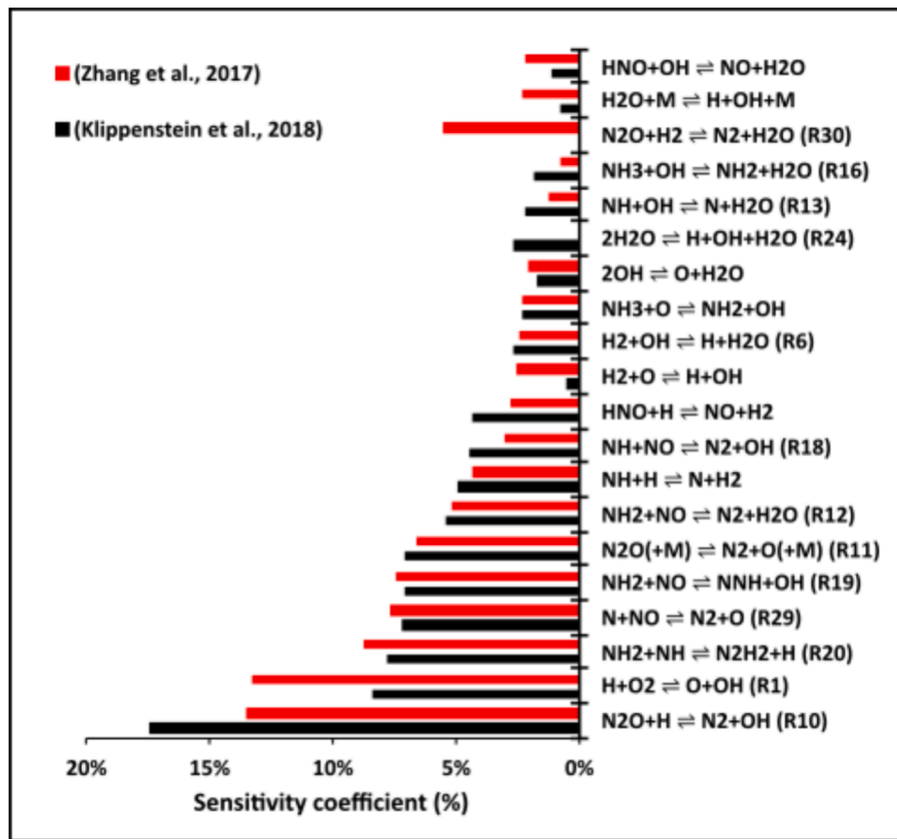


Fig. 18. Reactions with the largest negative local sensitivity coefficients for N<sub>2</sub>O mole fractions in 70/30 NH<sub>3</sub>/H<sub>2</sub> vol% premixed flame at  $\phi = 1$  in the Klippenstein and Zhang kinetic models.

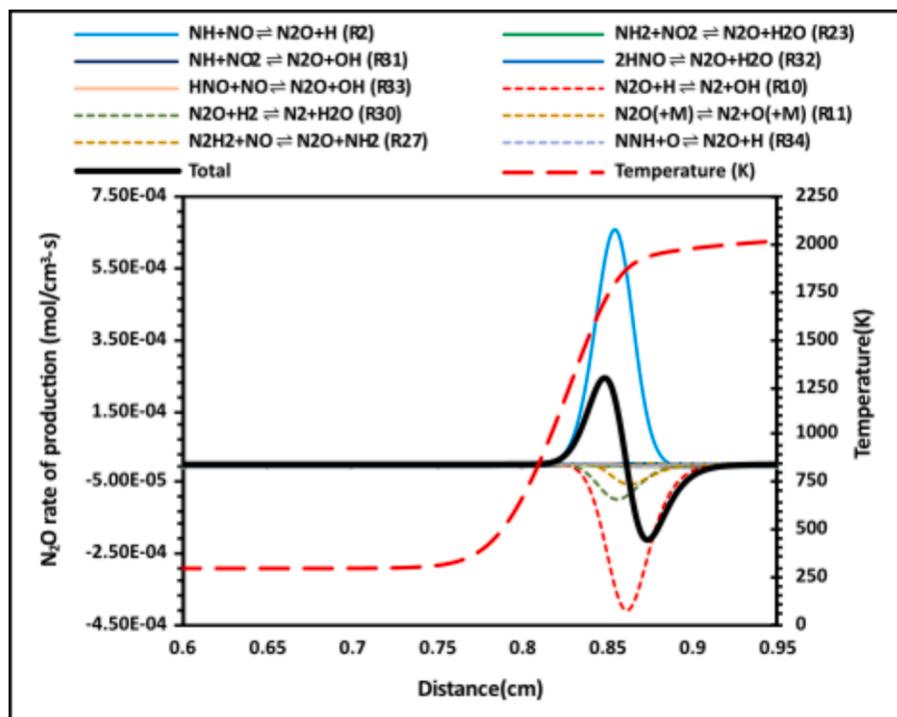


Fig. 19. The rate of production/consumption of N<sub>2</sub>O in 70/30 vol% NH<sub>3</sub>/H<sub>2</sub> mixture at stoichiometric conditions by the model of Zhang.

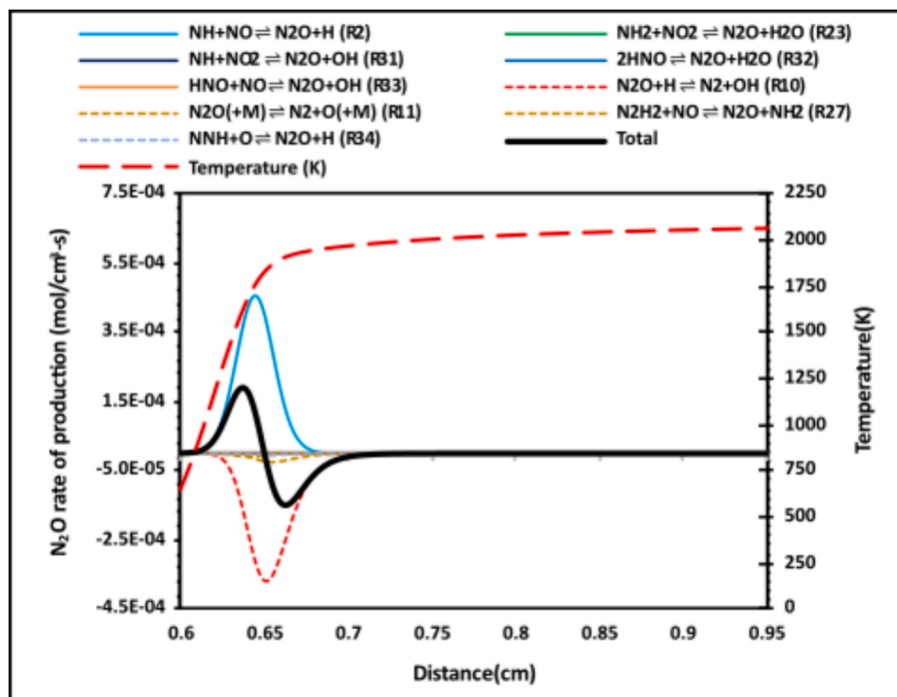


Fig. 20. The  $\text{N}_2\text{O}$  rate of production/consumption in 70/30 vol%  $\text{NH}_3/\text{H}_2$  mixture at stoichiometric conditions by the Klippenstein kinetic model.

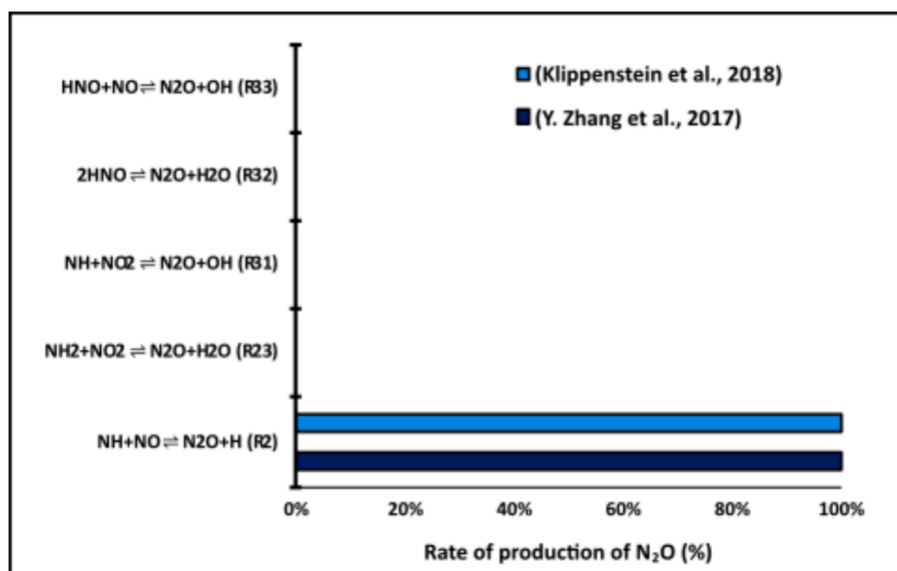


Fig. 21. Rate of production at stoichiometric conditions estimated by the Zhang and Klippenstein kinetic models.

for reaction R10 (Klippenstein: 17 %, Zhang: 14 %), making this reaction the most significant in slowing  $\text{N}_2\text{O}$  formation. Additionally, reactions R11 and R12 display slightly higher values in the Klippenstein model compared to the Zhang model. In contrast, the Zhang kinetic model predicts higher negative sensitivity trends for the reaction steps R1, R19, R20,  $\text{N} + \text{NO} \rightleftharpoons \text{N}_2 + \text{O}$  (R29) than the Klippenstein model. Furthermore, the reaction step  $\text{N}_2\text{O} + \text{H}_2 \rightleftharpoons \text{N}_2 + \text{H}_2\text{O}$  (R30), which contributes to an increase in the  $\text{N}_2\text{O}$  consumption in the Zhang mechanism, does not have an effect on the Klippenstein mechanism. Instead, Klippenstein's model emphasizes the role of reaction step R24 in inhibiting  $\text{N}_2\text{O}$  formation.

Fig. 19 and Fig. 20 showcase the rate distribution for the formation and consumption of  $\text{N}_2\text{O}$  across the computational domain, as

determined by the kinetic reaction mechanisms of Zhang and Klippenstein, respectively. The Figures clearly demonstrate that the total concentration of  $\text{N}_2\text{O}$  initially rises, mainly due to the  $\text{N}_2\text{O}$ -forming reaction R2. Other reactions, such as  $\text{R23}$ ,  $\text{NH} + \text{NO}_2 \rightleftharpoons \text{N}_2\text{O} + \text{OH}$  (R31),  $2\text{HNO} \rightleftharpoons \text{N}_2\text{O} + \text{H}_2\text{O}$  (R32), and  $\text{HNO} + \text{NO} \rightleftharpoons \text{N}_2\text{O} + \text{OH}$  (R33), contribute less significantly to the increase of the  $\text{N}_2\text{O}$  mole fraction in both models (see Fig. 21). Unlike R2, which has a low activation energy R32, and R33 exhibit slower reaction rates due to higher activation energies, acting as barriers to  $\text{N}_2\text{O}$  formation. This occurs because the reactants are consumed more readily by other reactions that require lower activation energy. This pattern is evident in both kinetic models, as illustrated in Fig. 19. On the other hand, the reduction in the total  $\text{N}_2\text{O}$  concentration is due to the consumption effects of reactions R10, R11, R27, R30, and

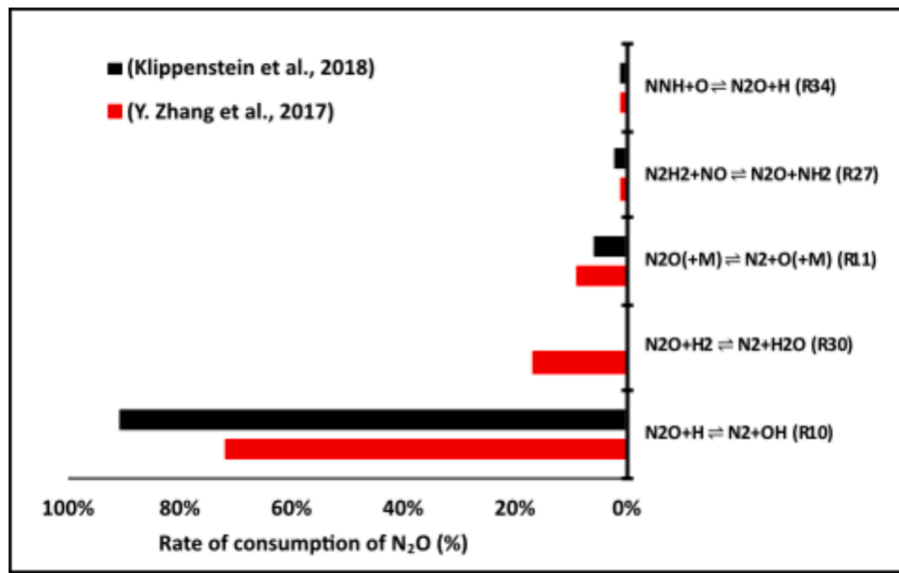


Fig. 22. Rate of consumption at stoichiometric conditions estimated by the Zhang and Klippenstein kinetic models.

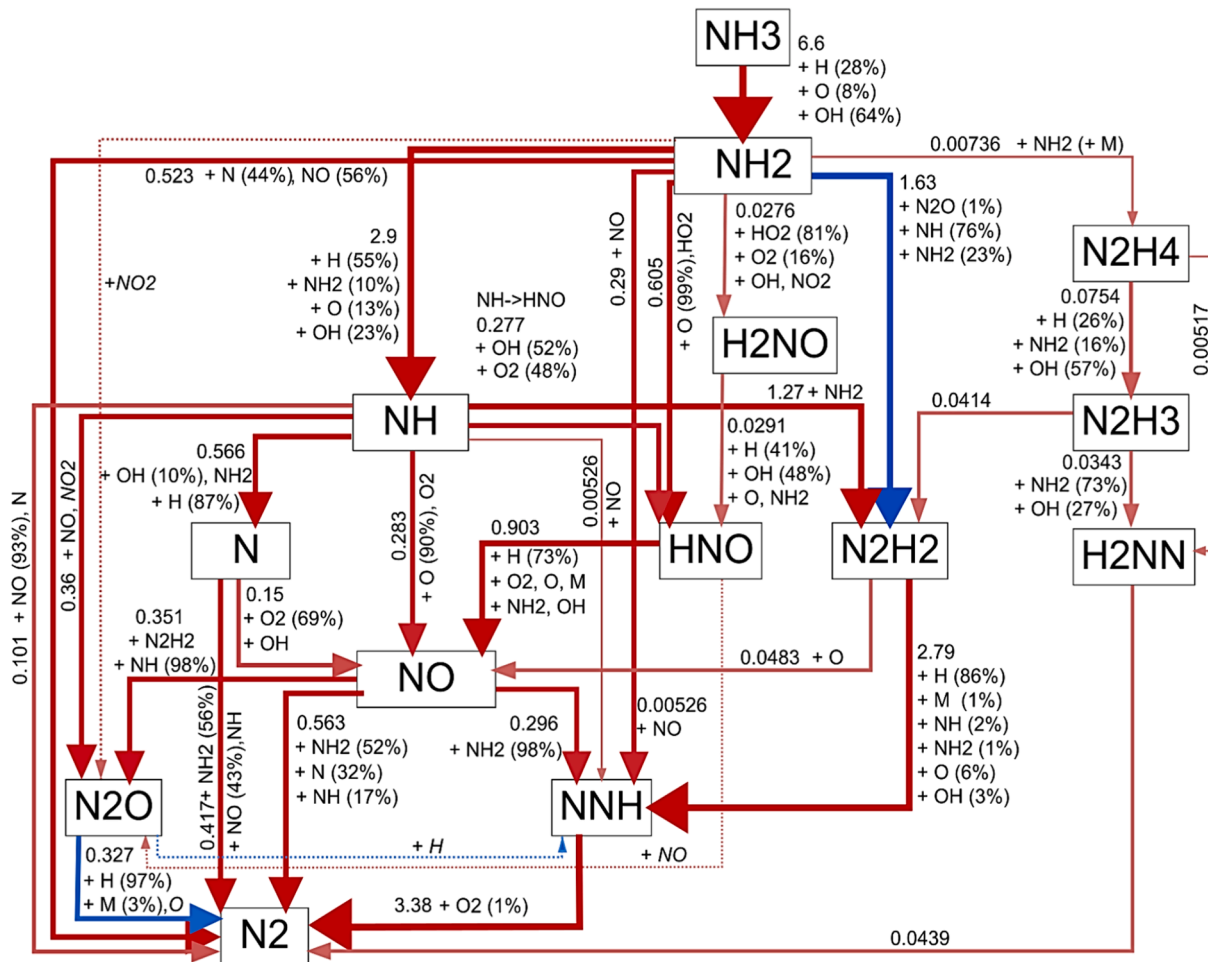


Fig. 23. Chemical reaction pathways of  $\text{N}_2\text{O}$  formation/consumption at flame zone ( $T = 1619 \text{ K}$ ) and at  $\phi = 1$  predicted by the Klippenstein model. Arrow lines refer to chemical transformations, percentages (%) refer to the contribution of reactants to the transformation, numbers stand for the net reaction rate in  $\text{kmol/m}^3\text{s}$ , which is also visualized by line thickness.

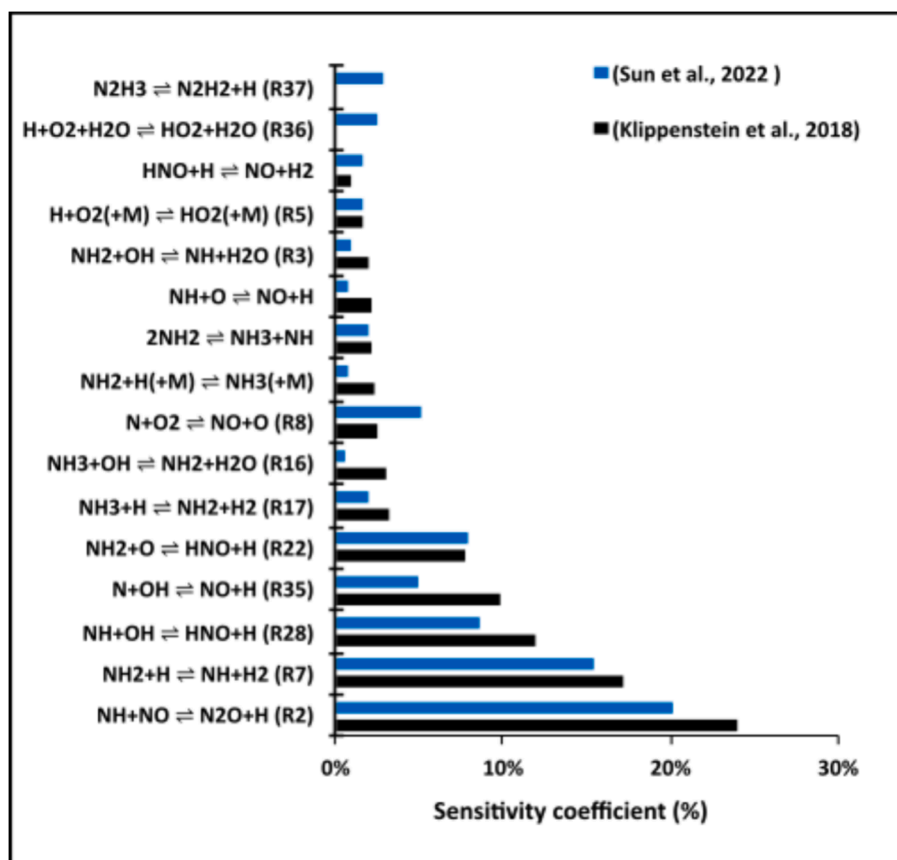


Fig. 24. Reactions with the largest positive local sensitivity coefficients for  $N_2O$  mole fraction in 70  $NH_3$ /30  $H_2$  vol% premixed flame at  $\phi = 1.4$  in the Klippenstein and Sun kinetic models.

$NNH + O \rightleftharpoons N_2O + H$  (R34), as indicated in Fig. 22. Additionally, within the Zhang mechanism, reaction R30 is identified as the second most significant reaction in reducing  $N_2O$  formation, a reaction not observed in the Klippenstein mechanism (see Fig. 22). Furthermore, the rate of  $N_2O$  consumption in reaction R11 is higher in the Zhang model compared to the Klippenstein kinetic model, which, in contrast, predicts higher consumption rates for reactions R10 and R27.

Fig. 19 and Fig. 20 also show that the Klippenstein model stabilizes the flame closer to the burner, with a flame speed of 27.62 cm/s, compared to the Zhang model, which has a flame speed of 21.38 cm/s [82].

Fig. 23 illustrates the main pathways for the formation/ consumption of  $N_2O$  in terms of net reaction rate in the reaction zone where  $T = 1619$  K. As shown in the figure, formation of  $N_2O$  from NO occurs mainly (in 98 %) via reaction with NH radicals according to R2, as well as dashed pathways:  $NH_2 + NO_2 \rightleftharpoons N_2O + H_2O$  (R23) and  $HNO + NO \rightleftharpoons N_2O + OH$  (R33) also contribute to the formation of  $N_2O$ . On the other hand, the chemical pathways for the consumption of  $N_2O$  shows that  $N_2O$  decomposes almost exclusively (in 99 %) into  $N_2$  by reacting with H atoms in reaction R10 ( $\sim 0.97 \times 99\% = 96\%$ ) and by unimolecular decay in R11 ( $\sim 3\%$ ), whereas the remaining 1 % of  $N_2O$  is consumed by reaction R26.

### 3.3. Rich flame conditions

The mole fraction of  $N_2O$  starts decreasing when the equivalence ratio increases to rich conditions and become close to zero at  $\phi = 1.4$  (see Fig. 3). In the rich range, the uncertainty values are large and exceed the experimental values of  $N_2O$ . This leads to statistically insignificant differences between simulations and experiments, falling within the expected variability of the measurements. The high magnitude of

uncertainty in rich conditions reflect the accuracy of most of the tested mechanisms in this range, as most of the kinetic models have a normalised error of less than 1. This indicates that the  $N_2O$  mole fractions predicted by most of the simulated kinetic models are within the error range of the experimental measurements. At an equivalence ratio ( $\phi$ ) of 1.4, the kinetic model of Klippenstein, which predicted the  $N_2O$  mole fraction, was within the error range (normalised error = 0.025). At the same time, several reaction mechanisms showed superior performance in the estimation of  $N_2O$ , such as the Sun kinetic model [47], which recorded a normalised error of 0.001 based on the experimental measurements, considering uncertainty. Therefore, both Klippenstein's and Sun's kinetic reaction mechanisms will be analysed in terms of sensitivity and rate of formation/consumption of  $N_2O$  to examine the reasons behind their discrepancies under these conditions.

As shown in Fig. 24, the  $N_2O$  mole fraction can be extremely boosted by the action of the reactions R2, R7, R22, R28 and  $N + OH \rightleftharpoons NO + H$  (R35). These Reactions are responsible for increasing the system reactivity by increasing the H and NH pools. It should be highlighted that reaction steps  $H + O_2 + H_2O \rightleftharpoons HO_2 + H_2O$  (R36) and  $N_2H_3 \rightleftharpoons N_2H_2 + H$  (R37) have no influence on the  $N_2O$  mole fraction in the Klippenstein mechanism, while they play a role in promoting  $N_2O$  formation in the mechanism of Sun. Fig. 25 illustrates that reactions R1, R10, R11, R12, R19, R20, and R29 have a considerable effect on reducing the concentration of  $N_2O$  by consuming H, NO, and NH species. Although both kinetic models show nearly the same reactions which have positive/negative trends on  $N_2O$  concentration, the estimated figures in most cases are different for the two mechanisms.

As shown in Fig. 26 and Fig. 27, the increasing trend of the total  $N_2O$  can be explained by the increasing rate of the  $N_2O$  producing R2 reaction. Furthermore, both selected kinetic models give the same estimation for the  $N_2O$ -production rate of reaction R2 (see Fig. 28). Meanwhile,

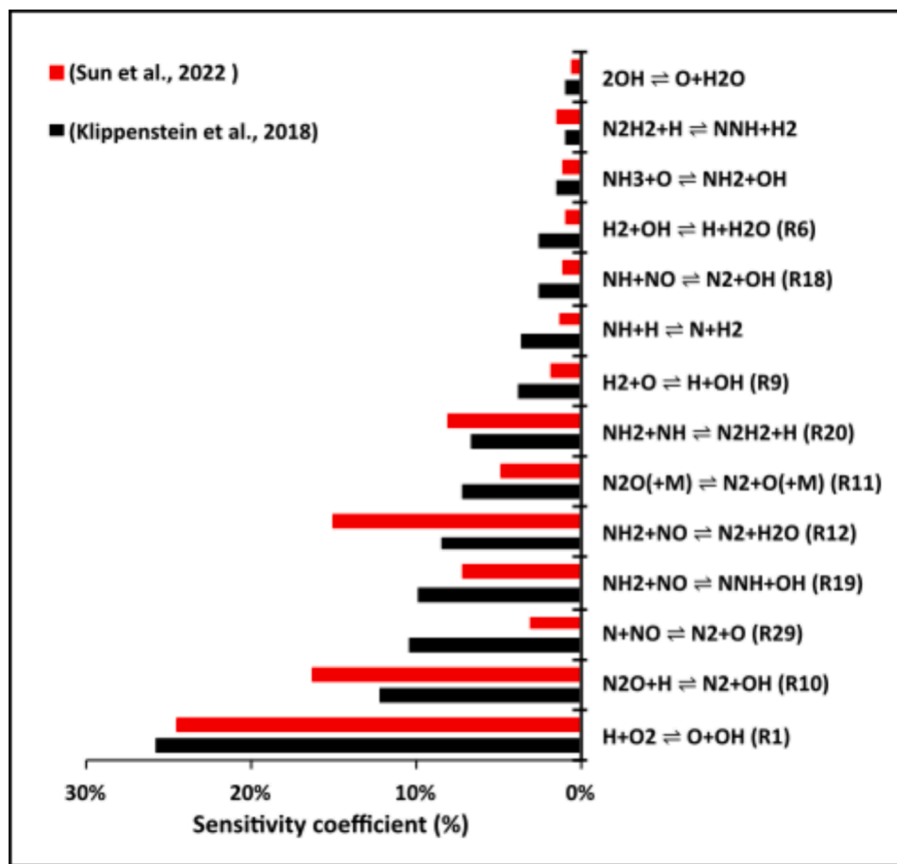


Fig. 25. Reactions with the largest negative local sensitivity coefficients for  $\text{N}_2\text{O}$  concentration in 70/30 vol%  $\text{NH}_3/\text{H}_2$  premixed flame at  $\varphi = 1.4$  in the Klippenstein and Sun kinetic models.

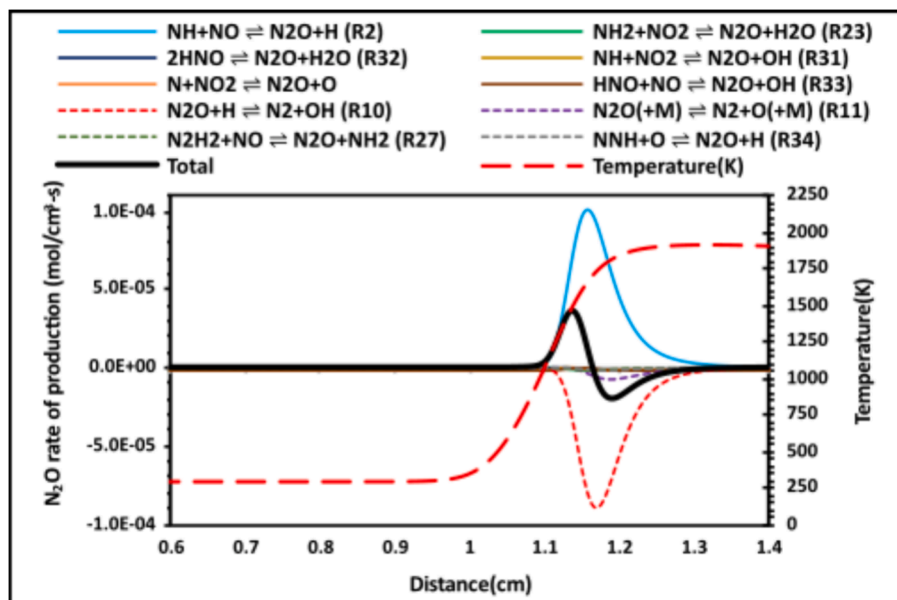


Fig. 26. The rate of production/consumption of  $\text{N}_2\text{O}$  in 70/30 vol%  $\text{NH}_3/\text{H}_2$  mixture at  $\varphi = 1.4$  estimated by the Sun kinetic model.

the peaking rates of  $\text{N}_2\text{O}$  consuming reactions R10 and R11 cause the sharp decrease in the total rate of  $\text{N}_2\text{O}$  concentration change. In addition, the rate of reaction R10 estimated by the Klippenstein kinetic model is slightly lower than that of Sun's reaction model (Fig. 29). Additionally, in rich conditions, despite both chosen models predicting nearly identical  $\text{N}_2\text{O}$  mole fractions, their predicted flame locations also

differ. The Klippenstein model predicts a flame location close to the inlet, suggesting a faster flame that stabilizes near the inlet, Fig. 27. In contrast, the Sun kinetic model indicates a slower flame, stabilizing further away from the inlet, Fig. 26.

It has been also observed the dominant role of the reaction R2 in increasing  $\text{N}_2\text{O}$  concentration, as well as the negative influence of

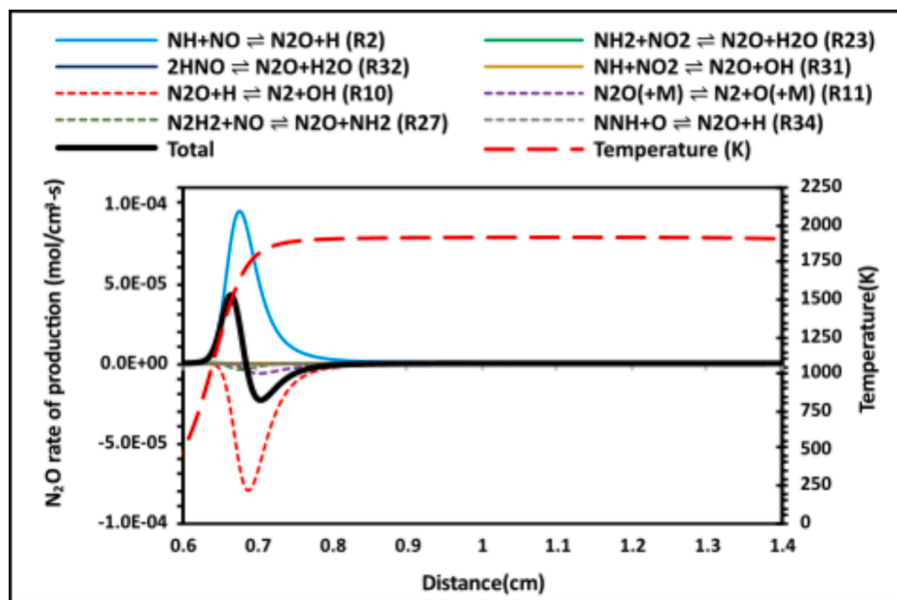


Fig. 27. The rate of production/consumption of  $N_2O$  in 70/30 vol%  $NH_3/H_2$  mixture at  $\phi = 1.4$  estimated by the Klippenstein model.

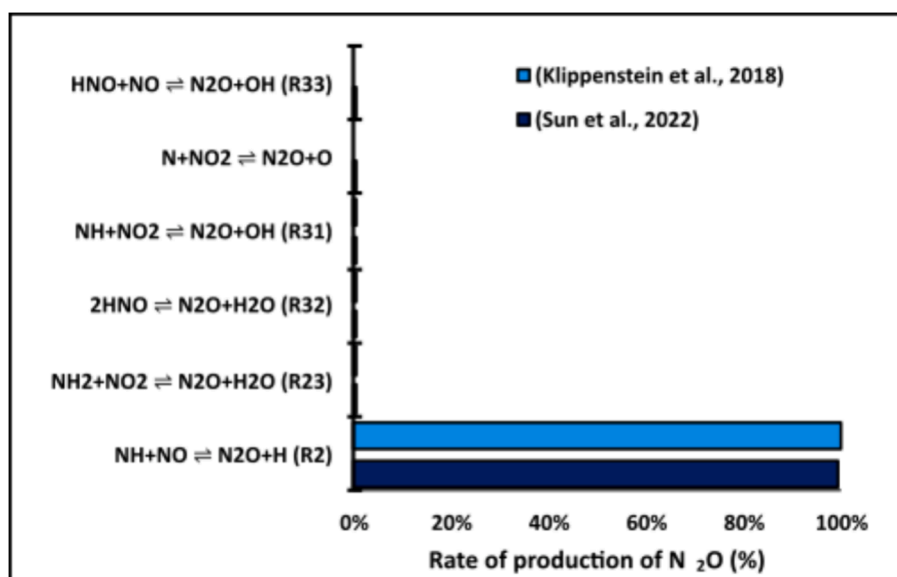


Fig. 28. Rate of production of  $N_2O$  at  $\phi = 1.4$  estimated by the Sun and Klippenstein kinetic models.

reactions R10 and R11 on the consumption of  $N_2O$  at rich conditions can be seen clearly in the pathway diagram in Fig. 30. The diagram shows the rates at 1478 K, which corresponds to the peak value of the total  $N_2O$  production rate. The pathway diagram shows that reaction  $NH + NO \rightleftharpoons N_2O + H$  (R2) accounts for about 98 % of the NO to  $N_2O$  transformation. Further, reaction  $N_2O + H \rightleftharpoons N_2 + OH$  (R10) is responsible in 98 % for the decomposition of  $N_2O$  to  $N_2$ , and the formation of  $N_2H_2$  from  $NH_2$  takes place in 23 % via reaction  $N_2H_2 + NO \rightleftharpoons N_2O + NH_2$  (R27) (see blue lines in Fig. 30).

From the previous sections, the investigation of the performance of tested kinetic models under different conditions of  $\phi$  by Klippenstein et al. [56], Glarborg et al. [50], Nakamura et al., Zhang et al. [18], and Sun et al. [47] demonstrated that kinematic differences among the mechanisms are the reason for the discrepancies in the predictions of the selected mechanisms. Additionally, the good performance of the kinetic models in specific conditions does not necessarily reflect their accuracy in predicting  $N_2O$  concentration.

The kinematic differences of the key reactions  $H_2 + N_2O \rightleftharpoons H_2O + N_2$ ,  $N_2O + M \rightleftharpoons N_2 + O + M$ , and  $N_2O + NH_2 \rightleftharpoons NO + N_2H_2$  among the kinetic models are clearly depicted in Fig. 31. According to the rate production analysis at stoichiometric conditions, the key reaction  $H_2 + N_2O \rightleftharpoons H_2O + N_2$  shows high values which significantly impacts  $N_2O$  consumption in the Zhang kinetic model. In contrast, it shows no effect in the kinetic models of Klippenstein, Nakamura, Sun, and Glarborg, indicating that the actual rate of this reaction is much lower as proved by Mulvihill et al. [88].

Similarly, the key reaction  $N_2O + NH_2 \rightleftharpoons NO + N_2H_2$  shows nearly the same rate constant in the tested kinetic models of Klippenstein, Zhang, Sun, and Glarborg, indicating a similar effect on the consumption of  $N_2O$  concentration. Only Nakamura's model shows a different trend. Meanwhile, the study by Cornell et al. [87] has shown that the rate constant of the reaction  $N_2O + NH_2 \rightleftharpoons NO + N_2H_2$  is likely overestimated in the tested kinetic models and should be reduced by a factor of 10, Fig. 31 (b).

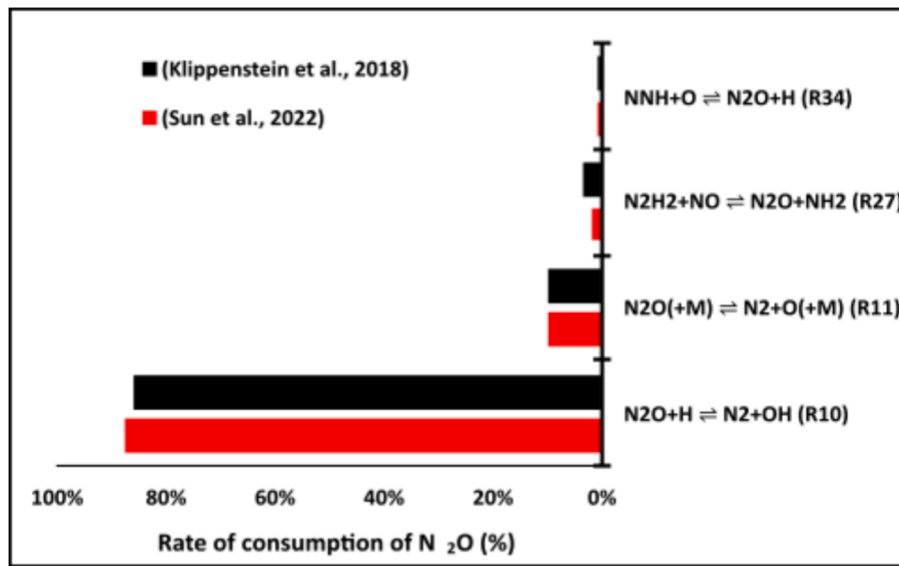


Fig. 29. Rate of consumption of N<sub>2</sub>O at  $\phi = 1.4$  estimated by the Sun and Klippenstein kinetic models.

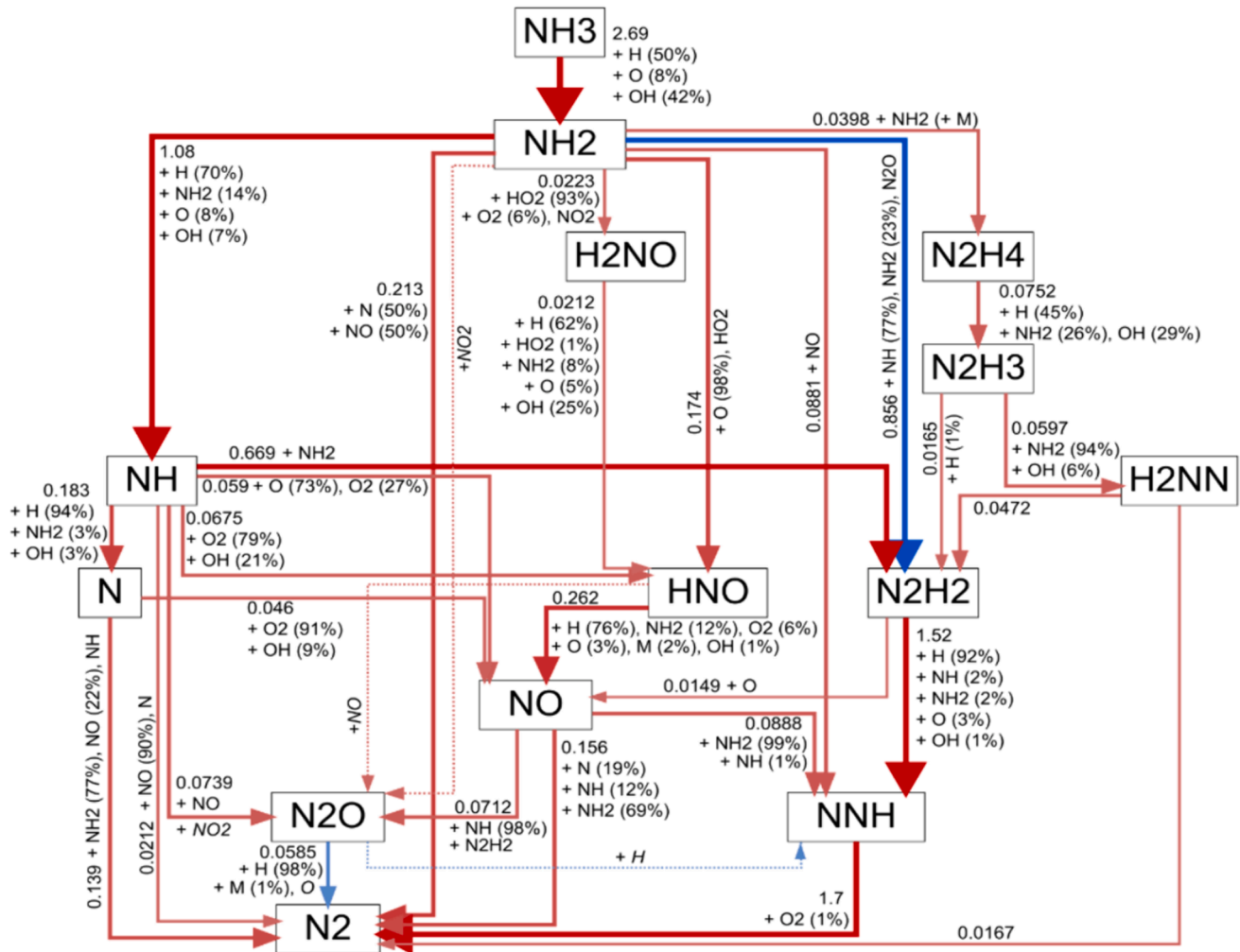


Fig. 30. Chemical reaction pathways of N<sub>2</sub>O formation/consumption at flame zone (T = 1478 K) and at  $\phi = 1.4$  predicted by the Klippenstein kinetic model. Arrow lines refer to chemical transformations, percentages (%) refer to the contribution of reactants to the transformation, numbers stand for the net reaction rate in kmol/m<sup>3</sup>s, which is also visualized by line thickness.

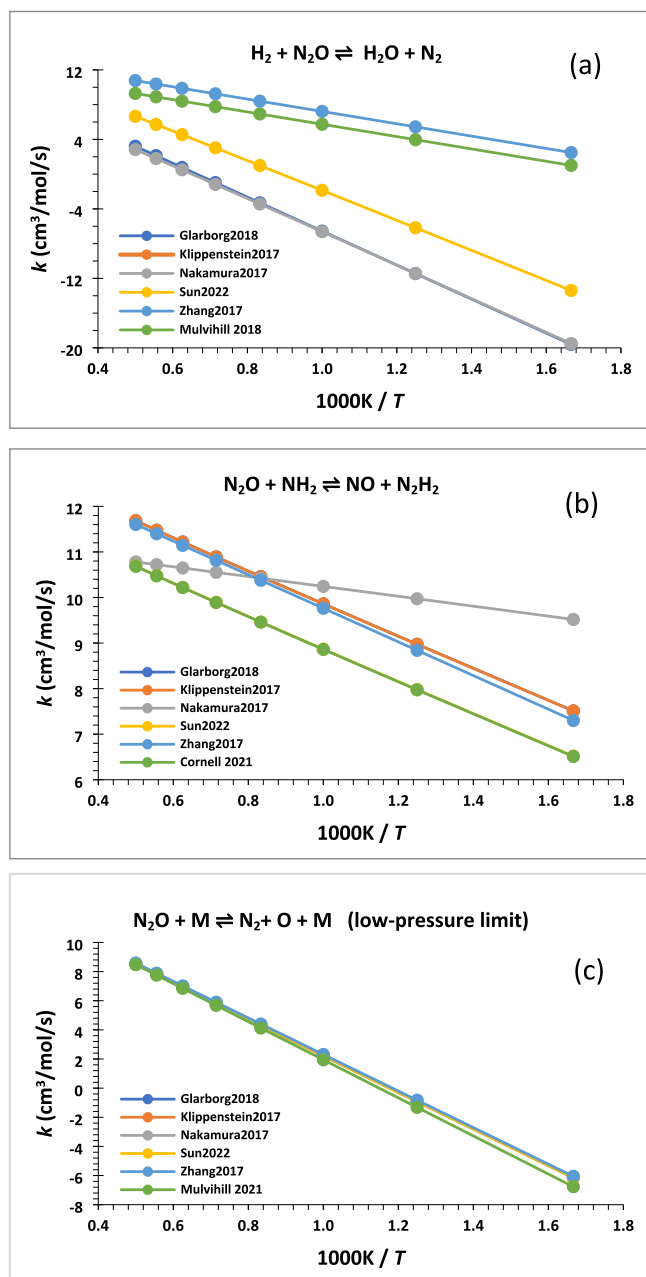


Fig. 31. Comparison of the rate constants for key reactions: (a)  $\text{H}_2 + \text{N}_2\text{O} \rightleftharpoons \text{H}_2\text{O} + \text{N}_2$ , (b)  $\text{N}_2\text{O} + \text{M} \rightleftharpoons \text{N}_2 + \text{O} + \text{M}$  (low-pressure limit), and (c)  $\text{N}_2\text{O} + \text{NH}_2 \rightleftharpoons \text{NO} + \text{N}_2\text{H}_2$ , according to recent studies and the adopted values from the tested kinetic models.

On the other hand, the rate constant for the key reaction  $\text{N}_2\text{O} + \text{M} \rightleftharpoons \text{N}_2 + \text{O} + \text{M}$ , as shown Fig. 31 (c), is the same among the tested mechanisms. However, according to the study by Mulvihill et al. [89], the rate constant should be corrected when the temperature exceeds 1500 K for an accurate estimation of  $\text{N}_2\text{O}$  concentration due to the reaction's major role in determining the  $\text{N}_2\text{O}$  concentration under given conditions.

#### 4. Conclusions

The present study numerically investigated the mole fraction of  $\text{N}_2\text{O}$  using 67 chemical kinetic mechanisms from the literature. The resulting numerical data were compared with experimental measurements from the literature using a normalised error approach, considering the uncertainty of the experimental measurements, to evaluate the

performance of the selected mechanisms in predicting  $\text{N}_2\text{O}$  mole fractions in a 70/30 vol%  $\text{NH}_3/\text{H}_2$  premixed flame. The study concludes that:

Based on the normalised error approach, most of the tested kinetic reaction mechanisms proved ineffective in predicting the  $\text{N}_2\text{O}$  mole fraction in the range of  $\phi = 0.57$ – $0.65$ , due to kinetic inhibition, which can lead to a failure to achieve ignition.

Klippenstein kinetic model generally predicts  $\text{N}_2\text{O}$  mole fractions accurately, but its performance deteriorates at very lean conditions ( $\phi = 0.57$ ).

The chemical reaction  $\text{NH} + \text{NO} \rightleftharpoons \text{N}_2\text{O} + \text{H}$  significantly contributes to the formation of  $\text{N}_2\text{O}$  under all tested conditions. The reduction of  $\text{N}_2\text{O}$  is primarily controlled by the reactions  $\text{N}_2\text{O} + \text{H} \rightleftharpoons \text{N}_2 + \text{OH}$ , and  $\text{N}_2\text{O} (+\text{M}) \rightleftharpoons \text{N}_2 + \text{O} (+\text{M})$ , which exhibit a dominant role across all equivalence ratios.

For local conditions of  $\phi$ , the Nakamura mechanism shows good performance at 0.6, while the Zhang and Sun kinetic models demonstrate superior performance at stoichiometric and rich conditions, where the normalised errors of the aforementioned kinetic models were less than 1.

Lastly, this study's findings focused on 70/30 vol%  $\text{NH}_3/\text{H}_2$  premixed flames at atmospheric conditions providing specific insights while also highlighting the limitations in assessing the generality of kinetic mechanisms. The majority of the evaluated models were developed for specific conditions, which limits their direct applicability to other  $\text{NH}_3/\text{H}_2$  blends due to variations in  $\text{N}_2\text{O}$  sub-mechanisms. A more comprehensive evaluation of these mechanisms' generality and predictive accuracy would require a broader dataset covering various  $\text{NH}_3/\text{H}_2$  blends and combustion systems. However, the current lack of such extensive experimental data, especially for the 70/30 vol%  $\text{NH}_3/\text{H}_2$  blend, means that the broader applicability of these mechanisms in predicting  $\text{N}_2\text{O}$  mole fraction remains an open question. Therefore, the current study represents a significant step in understanding these mechanisms under specific conditions and underscores the need for further research to fully explore and validate their generality across diverse combustion environments.

#### CRedit authorship contribution statement

**A. Alnasif:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **J. Jójka:** Writing – review & editing, Validation, Software, Resources, Data curation. **S. Mashruk:** Writing – review & editing, Visualization, Resources, Data curation. **T. Nagy:** Writing – review & editing, Visualization, Validation, Data curation. **A. Valera-Medina:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Investigation, Funding acquisition.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on request.

#### Acknowledgments

The authors would like to express their gratitude for the support received from EPSRC through the SAFE-AGT Pilot project (No. EP/T009314/1), the Green Ammonia Thermal Propulsion MarNH3 project

(No. EP/W016656/1), and Project AMBURN with funding from the Department for Energy Security & Net Zero (DESNZ) under award (No. IFS2-06-FLO). Additionally, Joanna Jójka acknowledges the funding received for her doctoral fellowship from the National Science Centre, Poland, under grant (No. UMO-2019/32/T/ST8/00265). Ali Alnasif also extends his thanks to Al-Furat Al-Awsat Technical University (ATU) for financially supporting his PhD studies in the U.K. Furthermore, T. Nagy is thankful for the support provided by the FK134332 NKFIH grant.

## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2024.131897>.

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